# (Not) Anything Goes: The syntax and diachrony of the German *gehen*-middle

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This paper offers a syntactic and diachronic analysis of the German [*gehen*+*zu*-INFINITIVE] construction. Previously misclassified as a periphrastic anticausative or conflated with canonical modal passives, the configuration represents a structurally distinct periphrastic realization of the middle voice. Synchronically, modal *gehen* operates as a functional raising head that strictly selects a voiceless, eventive *v*P and a resultative configuration — a selectional restriction that directly accounts for the construction's rigid lexical-semantic preference for Accomplishments. The raising head carries a formal edge feature forcing internal argument raising, which feeds dispositional evaluation at the conceptual-intentional interface. The resulting feature valuation simultaneously bleeds temporal auxiliary insertion and triggers post-syntactic *zu*-insertion as a morphotactic repair. Historical corpus data spanning the 15th to 20th centuries show that the *gehen*-middle emerged rapidly across the 18th and 19th centuries via a two-stage grammaticalization pathway: the lexical motion verb first underwent a semantic shift toward possibility, extending analogically into clausal control infinitives; ambiguous control structures were then reanalysed as monoclausal raising configurations. This reanalysis stripped the infinitival complement of its clausal architecture, reducing it to a bare *v*P, and bleached the matrix verb of its thematic subject, producing the constrained functional raising head attested in Present-Day German.

**Keywords:** middles; grammaticalization; dispositions; unaccusativity; raising; control; *gehen*.

## 1 Introduction

German exhibits a set of constructions in which an apparently active infinitival complement after *zu* is used to express passive voice. Among the most widely discussed of these are modal passives formed with the copula *sein* ('to be'), which convey possibility or obligation (see Höhle 1978; Holl 2010; Abraham 2020). Far less understood is a related construction that combines a finite form of the lexical motion verb *gehen* ('to go') with a *zu*-infinitive. This configuration, illustrated in (1), involves the promotion of the internal argument to the subject position, where it receives nominative case, while the external argument is demoted.

(1) *Die Tür geht leicht aufzuschließen.*
the door goes easy open-to-close
'The door unlocks easily | the door is easy to unlock.'

Crucially, the construction imposes a dispositional property onto this promoted subject, yielding a strong modal interpretation of dynamic (im)possibility. In this respect, the configuration closely resembles a canonical Germanic middle. However, it conspicuously lacks the reflexive morphology (e.g., *sich* 'self') required to license middles in German.

To understand this construction, we must distinguish this monoclausal raising structure from a superficially similar biclausal control structure. As shown in (2), *gehen* can also operate as a semi-modal control verb evaluating a clausal subject.

(2) *[Tabus abzuschaffen] geht gar nicht so einfach.*
taboos off-to-abolish goes at.all not so
'It is really not that easy to abolish taboos.' (Der Tagesspiegel, 30.11.1999, DWDS)

In the control configuration, the internal argument (*Tabus*) remains trapped within the embedded infinitival clause and retains accusative case. Furthermore, there is no agreement between the plural internal argument and the singular matrix verb *geht*. The monoclausal raising configuration in 1, by contrast, requires strict structural coherence and categorically bars the projection of an external argument.

Notably, the *gehen*-periphrasis has evaded rigorous empirical and theoretical investigation to date. Previous synchronic treatments have either relegated the construction to the margins of modal passives (Holl 2010; Müller 2019; Abraham 2020) or misclassified it entirely as an *Unmöglichkeitsantikausativ* ('impossibility anticausative') inherently driven by negation (Cysouw 2023).

This investigation provides, to our knowledge, the first extensive syntactic and diachronic analysis of the [*gehen*+*zu*-INFINITIVE] configuration. To capture its modern distribution, we deploy a formal acceptability judgment task, demonstrating that *gehen* operates as a functional raising head that enforces detransitivization through unaccusative syntax. Raising *gehen* strictly selects a voiceless, eventive *v*P and a resultative configuration. It then forces the internal argument to raise via a purely formal edge feature, which feeds a dispositional evaluation at the conceptual-intentional interface. Simultaneously, its prevalued status for [T] bleeds the insertion of temporal auxiliaries.

We argue that [*gehen*+*zu*-INFINITIVE] represents the direct structural reflex of a specific diachronic trajectory. By tracing the construction's emergence between the eighteenth and nineteenth centuries, we map a formal grammaticalization pathway wherein a semantic shift expressing possibility triggered the analogical extension of *gehen* into clausal control structures. These structures were subsequently reanalysed as the highly constrained monoclausal raising configurations observed today.

Section 2 establishes the theoretical and empirical background for *gehen*, detransitivization, and *zu*-infinitives. Section 3 presents the methodology and results of our quantitative acceptability judgment task. In Section 4, we present a discussion of the empirical findings through a theoretical lens. Section 5 provides a step-by-step formal derivation of the modern raising configuration. Finally, Section 6 contains a diachronic investigation making use of historical corpus data in order to propose a timeline for the reanalysis of [*gehen*+*zu*-INFINITIVE] from a control to a raising structure. We conclude the findings in Section 7.

## 2 Background

The *gehen* + *zu*-infinitive construction presents a structural anomaly. On the surface, the construction resembles a modal *zu*-infinitival passive (3a). Semantically, it carries a strong dispositional modality, i.e., a dynamic (im)possibility reading characteristic of middles and tough movement. Crucially, however, it lacks the reflexive marker *sich* that is required to license German middles (3b).

(3) a. *Aber jetzt geht es nicht mehr zu ändern.*
but now goes it not more to change
'But now it cannot be changed anymore.'

b. *Der Tisch geht leicht (*sich) abzuwischen.*
the.NOM table goes easy (*REFL) off-to-wipe
'The table wipes easily / can be easily wiped clean.'

The most detailed description to date is provided by Cysouw (2023) who describes the construction as an *Unmöglichkeitsantikausativ* 'impossibility anticausative'. This label encompasses two core observations: i) [*gehen*+*zu*-INFINITIVE] involves an apparent loss of causative behaviour, i,e. promotion of the Theme as a subject; and ii) attested instances often involves negation, invoking a dispositional property of the subject that impedes the denoted action (4a). However, as we shall show, the construction is neither limited to impossibility (as evident in 4b,c) nor is it classifiable as an anticausative.

(4) a. *Dieses Glas geht nicht zu schneiden weil es Sicherheitsglas ist.*
this glass goes not to cut because it safety-glass is
'This glass cannot be cut because it's safety glass'

b. *Die temporäre Datei geht zu löschen, aber nicht die exe.*
the temporary file goes to delete but not the exe
'The temporary file can be deleted but not the exe.'

c. *Das Radio geht zu reparieren.*
the radio goes to repair
'The radio can be repaired' (see examples in Cysouw 2023)

If the construction were a true anticausative, its lack of entailed or semantic Agent should freely facilitate modification by *von selbst* (cf. Chierchia 2004). However, our own and consultant judgements consider this to be strongly ungrammatical, (5). Moreover, [*gehen*+*zu*-INFINITIVE] categorically elicits a *with little effort* reading, while the interpretation *at the slightest provocation* is totally infelicitous (see Pitteroff 2014; Ackema and Schoorlemmer 2015). Together, these two tests strongly indicate that [*gehen*+*zu*-INFINITIVE] aligns much more closely with canonical middles and that an Agent is at least active in the semantics. Consequently, the construction cannot be considered a periphrastic anticausative (contra Cysouw 2023).

(5) a. *Die Tür geht leicht (*von selbst) zu öffnen.*
The door goes easy (by itself) to open.INF
'The door can be opened easily (*by itself)'

Moreover, Cysouw (2023) provides a short list of attested verbs in the construction (6), yielded from a corpus search.

(6) **Attested verbs**
*ändern* 'to change', *bauen* 'to build', *beheben* 'to rectify', *löschen* 'to extinguish', *reparieren* 'to repair', *rezipieren* 'to reciprocate', *schneiden* 'to cut', *stopfen* 'to stuff', *verschließen* 'to lock' (Cysouw 2023)

The 12 attested examples provided by Cysouw (2023) are insufficient to draw any strong conclusions on the exact lexical-semantic or syntactic behaviour of *gehen*. However, a testable lexical-semantic tendency does emerge in Cysouw's data; the attested verbs in (6) strikingly appear limited to verbs denoting Accomplishments (Vendler 1957). If this tendency holds up, and coupled with the established modal disposition attributed to the Theme, *gehen* resembles most closely established medio-passive constructions, e.g., canonical middles and *lassen*-middles. This observation provides a clear and testable hypothesis that [*gehen*+*zu*-INFINITIVE] qualifies as a non-standard periphrastic middle. Indeed, our own and consultant judgements appear to support this hypothesis in that native-speakers appear to reject canonical achievements (7a) and statives with (7b) *gehen*; as discussed below, both are ruled out in middles.

(7) a. **Der Autoschlüssel geht nicht/leicht zu finden.*
The car-keys go not/easy to find

b. **Die Welt geht nicht zu kennen.*
the world goes not to know

## 2.1 Voice Architecture and the Middle Voice

To formally evaluate this hypothesis, we assume a standard VoiceP architecture (Kratzer 1996; Schäfer 2008; Alexiadou et al. 2015) where transitivity alternations depend entirely on the projection of Voice and its featural content. Passives project an expletive active Voice lacking a specifier but retaining an active Agent -role. Detransitivized structures lacking an active Agent divide into two structural camps. Reflexive middles and marked anticausatives project a non-thematic Voice head with a formal [D] feature, forcing the insertion of the expletive *sich* in its specifier. Pure unaccusatives, however, lack the Voice projection entirely.

(8) a. **Active thematic Voice:** Transitives
$[_{\text{VoiceP}}\text{DP}_{\text{EXT}}[\text{Voice}\{+\text{AGENT}, \text{D}\}[_{v\text{P}}\ldots]]]$

b. **Non-active thematic Voice:** Passives
$[_{\text{VoiceP}}\text{Voice}_{\{+\text{Agent},\emptyset\}}[_{v\text{P}}\ldots]]$

c. **Expletive thematic voice**: Reflexive anticausatives, Middles
$[_{\text{VoiceP}}\textit{sich}[\text{Voice}_{\{\emptyset,\text{D}\}}[_{v\text{P}}\ldots]]]$

d. **Voiceless:**Pure Unaccusatives, raising verbs, e.g., *scheinen*
$[_{v\text{P}}\ldots]$ (cf. Schäfer 2008; Alexiadou et al. 2015)

The middle voice represents a syntactic and semantic configuration between passives and anticausatives. In Germanic languages, middles behave syntactically like unaccusative anticausatives in that they demonstrably lack a syntactically active Agent (Hale and Keyser 1987; Roberts 1987; Fagan 1992; Ackema and Schoorlemmer 1995, 2015; Steinbach 2002; Lekakou 2005; Schäfer 2008; Pitteroff 2014). Thus, they fail the standard tests for agentivity, which a passive passes: they reject *von*-phrases, agent-oriented adverbs, and control into purpose clauses. In German, canonical middles and reflexive marked anticausatives are syntactically identical, as both require *sich* (Schäfer 2008), e.g., *sich öffnen* 'to open', but semantically distinct. Nonetheless, like *gehen*, a middle fails the *von selbst* test (Chierchia 2004; Schäfer 2008). The restrictions are shown below (9):[1]

(9) *Das Buch liest sich leicht (*von dem Mädchen) (*absichtlich) (*um etwas zu lernen) (*by itself).*
the.NOM book.NOM reads REFL easily (*by the girl) (*intentionally) (*in.order something to learn)
'The book reads easily (*by the girl) (*intentionally) (*to learn something) (*by itself).'

German middles, in contrast, do involve a semantically active Agent at the Conceptual-Intentional (C/I) interface (Lekakou 2005; Schäfer 2008; Pitteroff 2014). Because of this conceptual agent, middles license instrumental adjuncts and *für*-phrases which associate with the Agent (10).

(10) a. *Das Buch liest sich gut (mit einem Kindle/ für Kinder).*
the book reads REFL well (with a Kindle/ for children)
'The book reads well (with a Kindle / for children).'

1. However, note that at least some speakers seem to accept modification by *von selbst* in middles.

More evidence of the presence of a semantically active Agent comes from examples that are ambiguous with reflexive marked anticausatives, i.e., those anticausatives which necessarily merge an expletive *sich* in the external argument position (Schäfer 2008). In such cases, the Agent permits a 'little effort'-reading (cf. Ackema and Schoorlemmer 2015; Pitteroff 2014) (see 11).

(11) *Die Tür öffnet sich leicht*
the door opens REFL easily

Middle Reading: 'The door opens easily.' - with little effort
Anticausative Reading: 'The door opens easily.' - at the slightest provocation

Canonical middles show notable lexical-semantic constraints. First, they encode a modal dispositional reading (much like [*gehen*+*zu*-INFINITIVE]) ascribing a property to the promoted internal argument that facilitates or impedes the event (Roberts 1987; Fagan 1992; Ackema and Schoorlemmer 1994, 1995, 2015; Steinbach 2002; Lekakou 2005). Moreover, middles impose strict restrictions on Vendlerian (1957) verb classes. While Germanic middles readily permit Accomplishments and Activities, they are incompatible with Achievements and Statives (Roberts 1987; Fagan 1992; Ackema and Schoorlemmer 1994, 2015; Lekakou 2005).

To satisfy the Middle's modification requirement, i.e., the addition of a facility adverb (see Condoravdi 1989), the promoted subject must function as an Undergoer or Theme (Ackema and Neeleman 2004; Ackema and Schoorlemmer 1995, 2015; Lekakou 2005) within a durative aspectual structure (cf. Fagan 1992). This durativity allows the progression of the event to be mapped onto the subject (Condoravdi 1989); this is unproblematic for Accomplishments and Activities. Achievements, however, denote an instantaneous change of state and cannot provide the aspectual progression required for this mapping. Statives also lack the dynamic progression necessary for the correct temporal mapping (see discussion in Lekakou 2005:164–172). Moreover, following Vendler (1957) and Dowty (1979), stative predicates are defined by their lack of dynamicity and internal control. This clashes with a reliance in middle formation for the detransitivized verb to possess an agentive logical subject (Roberts 1987; Ackema and Schoorlemmer 1994). Since statives lack Agents, they cannot entail the necessary Agent in the semantics (cf. Condoravdi 1989; Schäfer 2008).

## 2.2 Modal Infinitives and Restructuring in German

### Modal Infinitives

German hosts a range of constructions which necessitate a *zu*-infinitive. From a structral perspective [*gehen*+*zu*-INFINITIVE] resembles a range of very well-documented infinitival detransitivation strategies known as "modal infinitives" (Brinkmann 1962) or "modal passives"(Höhle 1978), such as *sein* 'be' + [*zu*-INFINITIVE] (12a) or lexical verbs such as *bleiben* 'stay' (12b) or *stehen* 'stand' (see also work by Haiden 2005; Holl 2010; Müller 2019; Abraham 2020) Syntactically, these constructions form obligatorily coherent verb clusters that strongly resist extraposition (Haider 2010; Holl 2010).

(12) a. *Das Ergebnis bleibt abzuwarten.*
The result remains off-to-wait
'The result must be waited for.'

b. *Die Zugfenster sind nachts zu schließen.*
The train-windows are night-GEN to close-INF
'The train's windows must be closed at night.'

A central theoretical debate in the literature concerns the exact locus of passivization in these structures. These break into several distinct groups: i) the infinitival marker *zu* itself acts as a passivizing element blocking Merge of an agentive External Argument (for different mechanisms see Haiden 2005; Haider 2010; Holl 2010), ii) the function of *zu* is orthogonal to passivization, and the passive reading is a result of the syntactic configuration (see also Demske-Neumann 1994) in the infinitive, e.g., a restructured and truncated VP (Wurmbrand 1998, 2001) or a specifierless VoiceP (Wurmbrand 2015), or iii) *zu* is not a passivizer; passivization results entirely from the unaccusativity of the copula *sein* (Abraham 2020).

Moreover, *sein+zu* infinitives are amenable to both a deontic and a dispositional (dynamic) possibility modality, i.e., *müssen* 'must' vs *können* 'can' (Holl 2010). In the classic view of (covert) modality in such contexts, the deontic reading requires a syntactic agent, while dynamic possibility does not (Höhle 1978; von Stechow 1990; Bhatt 2006), as illustrated in (13); the latter is often understood as tough movement, which produces a middle reading (Steinbach 2002). If correct, the deontic version must select an agentive functional projection introducing a covert syntactic Agent in the infinitival domain, e.g., VoiceP. For the possibility reading, it follows that *sein* selects an agentless vP complement (Wurmbrand 2001).[2] This is the general position that we shall adopt given its established support in the literature of modality and theta-structure in modal passives.

(13) The distribution of *von*- and *für*-Phrases with *sein+zu*:

a. *Die Aufgaben sind (unbedingt) von allen/*für alle zu lösen.*
the.NOM tasks.NOM are (absolutely) by all.DAT/*for all.ACC ZU solve.INF
'The tasks (absolutely) must be solved by everyone/*for everyone.' **Deontic**

b. *Die Aufgaben sind (leicht) *von allen/für alle zu lösen.*
the.NOM tasks.NOM are (easily) *by all.DAT/for all.ACC ZU solve.INF
'The tasks can (easily) be solved *by everyone/for everyone.' **Possibility**

2. We note that Wurmbrand (2001) uses only the presence of an adjective to distinguish *easy-to-please* and modal passives, i.e., selection of an AP by lexical $\mathrm{v}_{sein}$ and selection of a VP by modal $\mathrm{VAUX}_{sein}$. She cannot, however, derive the different modalities or provide unambiguous data to differentiate a structural bifurcation for possibility readings. Indeed, following convincing arguments by (Haider 2010:309ff), modifiers *leicht, schwer* 'easy, tough' in *sein+zu* must be treated as adverbials, while *sein* selects a verbal complement.

(ex.59 Holl 2010:36, following Höhle 1978)

Therefore, we conclude that a deontic modal passive includes an agentive VoiceP layer, while the possibility reading of *sein+zu* lacks one.

### Restructuring

Moreover, modal infinitives like *sein*+zu or indeed [*gehen+zu*-INFINITIVE] must be understood in the context of Restructuring. A useful distinction can be drawn between Lexical Restructuring Infinitives (LRIs) and Functional Restructuring Infinitives (FRIs) (Wurmbrand 2001, 2004; Bobaljik and Wurmbrand 2005; Wurmbrand 2015). LRIs are exclusively governed by θ-assigning lexical verbs, e.g., *versuchen* 'try'; these structures involve control and can be active or passive. In (14), we see that with lexical restructuring coherence effects such as long scrambling of the embedded object are possible without cluster formation (in the so-called 'third construction', Wöllstein-Leisten 2001). The standard assumption is a PRO in spec,vP or rather VoiceP. The Agent of the restructuring verb controls the covert Agent of the embedded verb.[3]

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| (14) | *weil* | *er* | *den/*der* | *Traktor* | *versucht* | *hat* | [*t_OBJ* | *zu* | *reparieren*] |
| | since | he | the.ACC/*the.NOM | tractor | tried | has | | to | repair |

'since he tried to repair the tractor' (ex.5a, Bobaljik and Wurmbrand 2005:814)

In contrast, FRIs typically involve a non-θ-assigning functional head in the matrix domain, e.g., *scheinen* 'seem'. These verbs permit raising of the highest argument in the embedded infinitival domain, but do not necessitate Control.

| | | | | | | |
|---|---|---|---|---|---|---|
| (15) | [*Der* | *Fehler*$_i$] | *scheint* | $t_i$ | *zu* | *verschwinden.* |
| | the.NOM | error.NOM | seems | | ZU | disappear.INF |

'The error seems to disappear.' **subj-to-subj raising**

The categorical status of functional restructuring heads like *scheinen* is debated, analysed either as high functional operators in the clausal spine (Cinque 1999, 2006a; Wurmbrand 2001, 2004; Roberts 2010) or as unaccusative *v* heads selecting a reduced complement (Grewendorf 1989; Chomsky 2000, 2001; Davies and Dubinsky 2004; Sternefeld 2006; Haider 2010; Abraham 2020). Because both approaches fundamentally rely on the selection of a defective domain, we model functional restructuring via an unaccusative v head. This structural choice avoids the empirical difficulty of explaining why a high auxiliary would idiosyncratically force a *zu*-infinitive rather than a bare infinitive. The predicate's epistemic or modal content can instead be derived formally via a feature on v, e.g., [i:Mod *val*].

Moreover, LRIs and FRIs show different behaviour in terms of extraposition. Raising

3. The mechanism of Control is not important here, yet it has been modelled in terms of PRO (Chomsky 1981; Landau 2013), post-syntactic Obligatory Control (Wurmbrand 2001), pre-syntactic argument pooling (Haider 2010), or Voice matching (Wurmbrand 2015)

*scheinen* (Haider 1993; Wurmbrand 2001) and *sein+zu*(Demske-Neumann 1994; Wurmbrand 2001; Haider 2010) are both incompatible with extraposition (16a,b), requiring strict coherence of an apparent cluster VRAISING+*zu*+VINFINITIVE. However, control structures and particularly lexical restructuring verbs in long passives (17), the so-called 'third construction', do allow extraposition of the *zu* infinitive away from the promoted subject in the matrix clause (Wöllstein-Leisten 2001; Müller 2020; Haider 2010).[4]

(16) a. **dass Hans schien den Kuchen gegessen zu haben*
that John seemed the cake eaten to have.INF
'that John seemed to have eaten the cake' (ex. 116n Wurmbrand 2001:158)

b. **dass der Kuchen nicht ist zu essen*
that the.NOM cake.NOM not is zu eat.INF
'that the cake cannot/may not be eaten' (ex. 116o Wurmbrand 2001:158)

(17) *... dass [der Hund] vergessen wurde, [zu füttern].*
... that the dog.NOM forgotten was, to feed.
'that it was forgotten to feed the dog.' (ex.2c Haider 2010:285)

Therefore, we have established a range of metrics to elucidate the status of *gehen+zu* and inform any theoretical analysis.

## 3 Experiment

We conducted a quantitative acceptability judgment task for [*gehen+zu*-INFINITIVE] targeting the core diagnostic properties distinguishing raising, passive, and middle structures. We recruited 72 L1-German participants to rate 50 randomized stimuli (presented with context) on a 1–5 Likert scale. The stimuli tested various conditions placed on [*gehen+zu*-INFINITIVE] itself and compared it to other semantically or syntactically aligned structures, including *lassen*-middles, canonical *werden*-passives, and modal *zu*-infinitives with *sein*.

The experimental design specifically targeted the core diagnostic properties capable of differentiating raising, passive, and middle structures. These included structural restrictions at the TP and vP levels (periphrastic tense, extraposition, and dative/double object preservation) (Haider 1993, 2010; Wurmbrand 1999, 2001), semantic and syntactic agentive properties (*von*- vs. *für*-phrases, instrumental adjuncts) (Keyser and Roeper 1984; Roberts 1987; Baker et al. 1989; Fagan 1992; Ackema and Schoorlemmer 1995, 2015), and lexical-semantic aspectual restrictions (cf. Vendler 1957).

Linear Mixed-Effects Models were fitted with crossed random intercepts for participants and

4. We note that Wurmbrand (2001) is less accepting of this construction, yet we follow the majority position in the literature.

items.[5] Fixed effects were sum-coded for factorial designs; treatment coding was used for multi-level single factors, with canonical constructions as the reference level. Descriptive statistics for all conditions are visualized in the accompanying figures. Prose reports standardized means ($M_z$) alongside inferential model output $\beta$, *SE*, $z$, $p$).

## 3.1 Establishing relative acceptability thresholds

Non-standard constructions systematically suffer from prescriptive bias in formal tasks, resulting in deflated ratings (Schütze 1996; Cornips and Poletto 2005). Two structural anchors were established for interpreting gradient well-formedness (cf. Featherston 2005, 2007). The absolute experimental ceiling ($M = 4.63, M_z = 1.19$) was defined via grammatical canonical passive fillers ($n = 9$), while the floor ($M = 1.32, M_z = -1.04$) used ungrammatical controls ($n = 3$). The highest-rated *gehen* item ($M = 3.85, M_z = 0.64$) serves as the construction-specific ceiling ($C_{ns}$), revealing a prescriptive penalty of approximately 0.78 raw scale points ($\Delta_z = 0.55$) (Table 1).

**Table 1:** Relative Acceptability Thresholds by Construction Type

| **Tier** | **Canonical (Ceiling** $M_z = 1.19$**)** | **Non-Standard (Ceiling** $M_z = 0.64$**)** |
|---|---|---|
| Fully Licensed | $M_z \geq 0.65$ ($M \geq 3.90$) | $M_z \geq 0.40$ ($M \geq 3.50$) |
| Marginally Licensed | $-0.15 \leq M_z < 0.65$ | $-0.30 \leq M_z < 0.40$ |
| Rejection | $M_z < -0.15$ ($M < 2.60$) | $M_z < -0.30$ ($M < 2.50$) |

In all boxplots, dashed horizontal lines indicate standardized thresholds of acceptability: the absolute canonical ceiling (green) and floor (red), alongside the fully licensed threshold (blue) and the rejection floor (blue) for the *gehen*-middle. In Appendix A, we provide tables with the raw data for each test and a list of the experimental stimuli, alongside their raw and $z$-score standardized acceptability means.

## 3.2 Experimental Data: Results

### 3.2.1 Structural restrictions at TP and vP

#### Compound tense

We tested the acceptability of synthetic and periphrastic morphology for *gehen* compared to the canonical raising predicate *scheinen*, which is known to categorically resist compound tense morphology (Haider 1993, 2010; Wurmbrand 2001). The contrasts are given in (18–19).

5. Statistical modeling and data visualization were performed in Python 3 using the `statsmodels` library.

(18) a. *Die Journalistin schien gut informiert zu sein.*
The journalist seemed well informed to beINF
'The journalist seemed to be well informed.'

b. **Die Journalistin hat gut informiert zu sein geschienen.*
The journalist has well informed to be seemed.PTCP

(19) a. *Der Wagen ging nicht zu reparieren.*
The car went not to repair.INF
'The car could not be repaired.'

b. **Der Wagen ist nicht zu reparieren gegangen.*
The car is not to repair gone.PTCP

As shown in Figure 1, simple forms of both *scheinen* ($M_z = 1.39$) and *gehen* ($M_z = 0.42$) are within their licensed ranges. However, periphrastic forms are ungrammatical, with both *scheinen* ($M_z = -0.94$) and *gehen* ($M_z = -0.95$) clustering at the absolute experimental floor ($M_z = -1.04$).

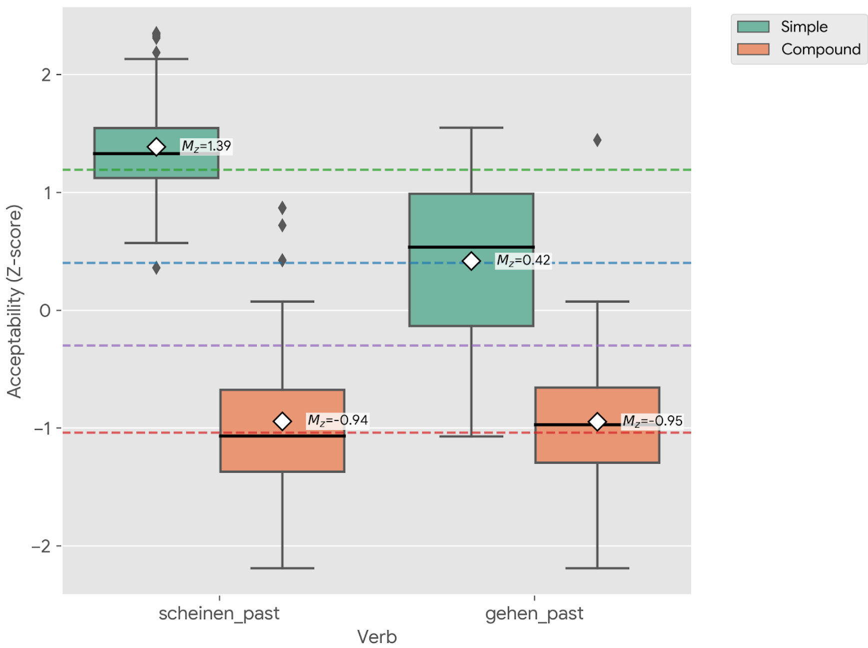



**Figure 1:** Distribution of acceptability (Z-scores) for simple and compound tense forms

An LMM revealed significant main effects for MORPHOLOGY ($\beta = 2.76$, $SE = 0.10$, $z = 26.41$, $p < .001$) and VERB ($\beta = 0.72$, $SE = 0.10$, $z = 6.85$, $p < .001$), alongside a significant interaction ($\beta = 1.38$, $SE = 0.21$, $z = 6.59$, $p < .001$). The prescriptive penalty against *gehen* is restricted entirely to simple tenses. In compound tenses, both verbs categorically collapse to the experimental floor and are statistically indistinguishable ($\beta = 0.03$, $SE = 0.12$, $z = 0.23$, $p = .815$). This confirms that *gehen* patterns with a functional raising head like *scheinen* (cf.

Wurmbrand 2001).

### Extraposition

The extraposition test distinguishes lexical heads, which permit rightward displacement of the *zu*-infinitive (Wöllstein-Leisten 2001; Müller 2020; Haider 2010), from functional restructuring heads, which require a coherent intraposed cluster (Haider 1993, 2010; Wurmbrand 2001). Extraposition of [*zu*-INFINITIVE] is definitively judged as ungrammatical ($M_z = -1.03$), falling at the experimental floor ($M_z = -1.04$), while the intraposed form ($M_z = 0.30$) falls close to the fully licensed range (20) (See Figure 2).

(20) *Louisa ist sehr glücklich ...*
('Louisa is very happy ...')
*dass das Spiel endlich *(zu installieren) geht (*zu installieren).*
that the game finally (to install) goes to install
'that the game finally can be installed.'

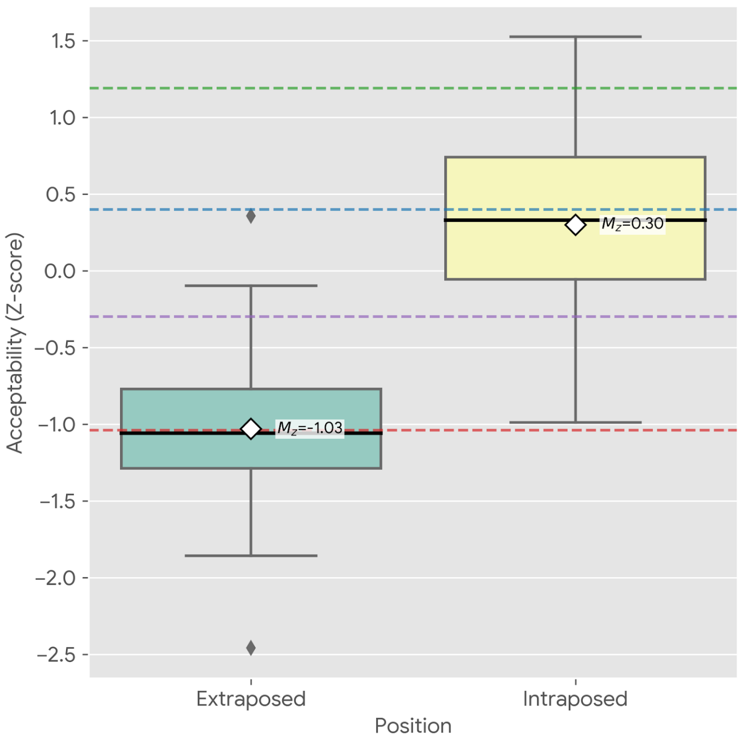



**Figure 2:** Distribution of standardized acceptability (Z-scores) for intraposed vs. extraposed infinitives with *gehen*

An LMM utilizing treatment coding (with the intraposed condition as the reference level) confirmed this collapse in acceptability to be highly significant ($\beta = -2.00$, $SE = 0.14$, $z = -14.30$, $p < .001$). This demonstrates that *gehen* behaves as a functional raising head that necessitates a coherent infinitive structure. Consequently, *gehen* behaves like the functional raising verb*scheinen* and modal passive [*sein*+*zu*-INFINITIVE]. This distinguishes *gehen* from lexical restructuring heads in long passives, which freely allow rightward displacement of the infinitival complement. This confirms that [*gehen*+*zu*-INFINITIVE] is a special type of raising structure.

### Datives and double arguments

Canonical middles resist dative arguments, whereas modal passives and tough-movement constructions permit them (Fagan 1992; Ackema and Schoorlemmer 1995; Steinbach 2002). We thus compared dative-assigning *helfen* ('help') across *sein* and *gehen* configurations to see if this constraint extends to [*gehen*+*zu*-INFINITIVE]. (21)

(21) *Ihm ist/*geht nicht zu helfen.*
him.DAT is/goes not to help.INF
'He cannot/should not be helped.'

As visualized in Figure 3, the *sein*-passive remains around the canonical ceiling ($M_z$ = 1.32), while *gehen* is categorically rejected, clustering at the absolute experimental floor ($M_z = -0.99$). An LMM utilising treatment coding (with [*sein*+*zu*-INFINITIVE] as the reference level) confirmed a total collapse in acceptability ($\beta = -3.43$, $SE = 0.11$, $z = -31.11$, $p < .001$).

However, there is a methodological confound in the *gehen* condition regarding the monotransitive stimulus *helfen* ('to help'). As established in Section 2, *gehen* prefers Accomplishment predicates. Since *helfen* is an Activity, it was likely subjected to a double penalization: a violation of an aspectual requirement and a potential structural violation. This confound cannot be resolved by substituting the verb, since German systematically lacks dative-assigning two-place Accomplishments. Dative resultatives appear limited to punctual Achievements (e.g., *begegnen* 'to meet', *entkommen* 'to escape', *auffallen* 'to stand out'), which are independently incompatible with middles and *gehen*, as we will show. Consequently, the monotransitive data alone cannot prove a "dative allergy".

Crucially, however, our ditransitive condition (built with ditransitive Accomplishment *schicken* 'send') successfully isolates a structural restriction against datives and bypasses any aspectual confound. While ditransitive predicates are grammatical with the modal [*sein*+*zu*-INFINITIVE]-passive (22), *gehen* rejects them (23). Indeed, the ditransitive *gehen* stimulus yielded one of the lowest ratings ($M_z = -1.14$).

(22) *Das Buch ist ihm/dem Eigentümer zurückzugeben/auszuhändigen.*
the book is him/the.DAT owner back-to-give/hand-to-over
'The book must be given/handed over to the owner.'

(23) **Die Email geht ihm nicht zu schicken.*
the email goes him not to send
Intended: 'The email cannot be sent to him.'

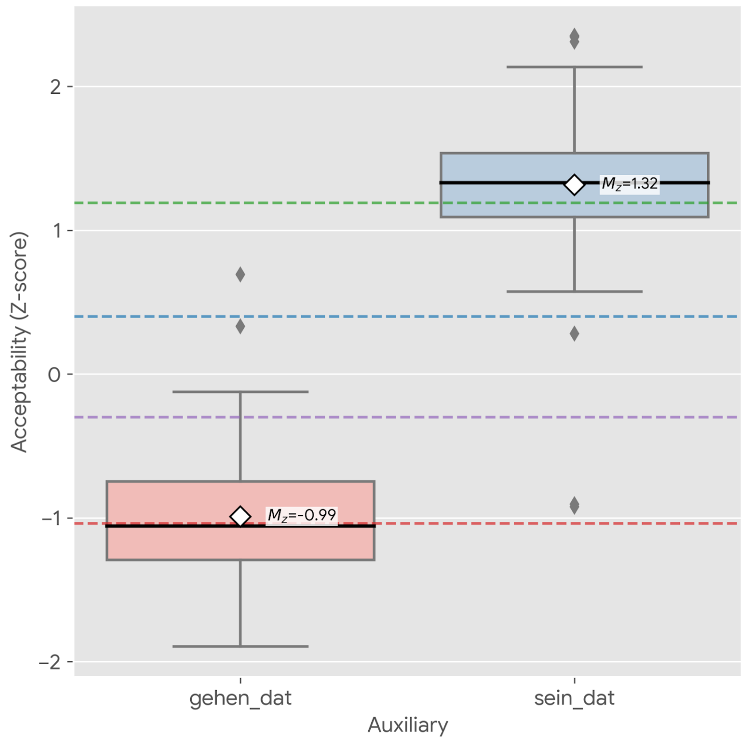



**Figure 3:** Acceptability ($Z$-scores) of *sein* vs. *gehen* with a dative-assigning verb *helfen*

#### 3.2.2 Semantic and syntactic agentive properties

In this subsection, we investigate the presence or absence of a covert syntactic Agent or an implicit semantic Agent. This is because passives are well known to pass diagnostics for a syntactially active agent, while canonical middles in Germanic pass only tests for a semantically active Agent (Roberts 1987; Baker et al. 1989; Fagan 1992; Steinbach 2002; Lekakou 2005; Ackema and Schoorlemmer 1995, 2015; Pitteroff 2014; Alexiadou et al. 2015).

#### Agentive *von*-phrases

Canonical passives (including modal *sein+zu*, see 24) unproblematically license agentive *von*-phrases, signalling a syntactically active but demoted external argument (Baker et al. 1989), yet canonical middles categorically reject them (Ackema and Schoorlemmer 2015). We tested the compatibility of *gehen* with a *von*-phrase against these established baselines to diagnose the syntactic status of its implicit Agent.

(24) *Das Ergebnis eines demokratischen Prozesses sei von allen zu respektieren*
the.NOM result.NOM a.GEN democratic.GEN process.GEN be.SBJV.3SG by all.DAT to respect.INF
'The result of a democratic process is to be respected by everyone.'
(DWDS, *Berliner Zeitung* 02.12.2004)

*gehen* with a *von*-phrase (26) is categorically rejected ($M_z = -0.67$), representing a massive structural penalty ($\Delta M_z = 1.31$) compared to baseline *gehen*-middles.The data is visualized in Figure 4.

(25) **Die Tür geht von Anna kaum aufzuschließen.*
the.NOM door.NOM goes by Anna barely up-to-lock
'The door can barely be unlocked by Anna.'

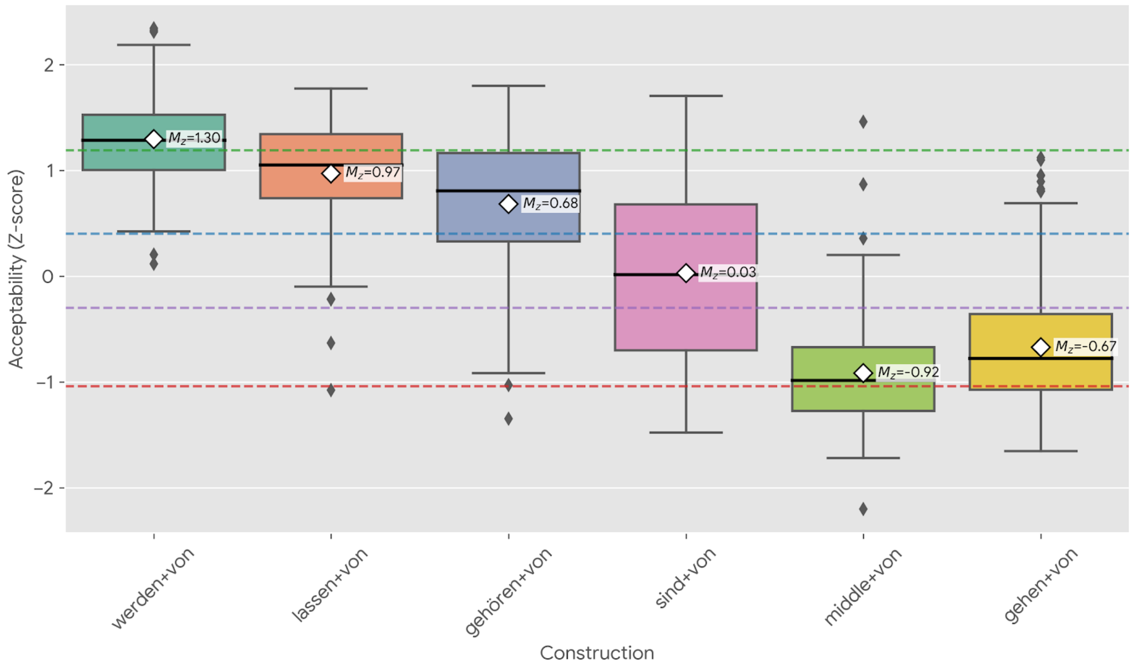



**Figure 4:** Distribution of acceptability ($Z$-scores) for agentive *von*-phrases across passive and middle constructions. The green dashed line marks the canonical ceiling ($M_z = 1.25$); the red dashed line indicates the experimental floor ($M_z = -1.04$). The blue dotted line indicates the threshold for fully licensed *gehen* items ($M_z = 0.40$), and the purple dash-dotted line marks its theoretical rejection threshold ($M_z = -0.30$).

An LMM utilizing treatment coding (with *gehen* set as the reference baseline) confirmed highly significant structural divergence from all passive structures. Canonical *werden*-passives ($\beta = 2.88$, $SE = 0.10$, $z = 30.25$, $p < .001$), *lassen*-passives ($\beta = 2.47$, $SE = 0.12$, $z = 20.51$, $p < .001$), *gehören*-passives ($\beta = 2.04$, $SE = 0.12$, $z = 16.93$, $p < .001$), and modal *sein+zu* passives ($\beta = 1.02$, $SE = 0.12$, $z = 8.50$, $p < .001$) all license explicit agents with significantly higher acceptability. Crucially, *gehen* patterns strictly with the agentless reflexive middle, which exhibits a similarly deep rejection ($\beta = -0.38$, $SE = 0.12$, $z = -3.15$, $p = .002$).

These data confirm that *gehen* lacks the functional architecture required to host a syntactically active external argument, pointing towards its classification as a purely dispositional, middle-voice construction.

### Implicit agents and *für*-phrases

While canonical middles categorically reject agentive *von*-phrases, they unproblematically license *für*-phrases. The felicity of a *für*-phrase reliably diagnoses the presence of an implicit Agent that is active in the semantics but absent from the narrow syntax (Hale and Keyser 1987; Fagan 1992; Ackema and Schoorlemmer 1995; Lekakou 2005; Schäfer 2008; Pitteroff 2014). To evaluate the status of the implicit Agent in *gehen*, we compared *für*- vs. *von*-phrase compatibility across *gehen*, canonical reflexive middles, and modal *sein+zu* passives.

As shown in Figure 5, a *für*-phrase raises *gehen* from categorical rejection with a *von*-phrase ($M_z = -0.67$) into the marginally acceptable zone ($M_z = -0.08$). Similar but stronger contrasts were observed in the canonical constructions. Canonical middles heavily preferred *für*-phrases ($M_z = 0.54$) over *von*-phrases ($M_z = -0.92$), and *sein+zu* structures operated at the canonical ceiling with *für*-phrases ($M_z = 1.23$) while degrading to marginality with *von*-phrases ($M_z = 0.03$).

We note that a potential ameliorating factor for the *für*-phrase with *sein*+*zu*, and thus penalty against *von*, may have been the nature of the predicate. [*sein*+*zu*-INFINITIVE] is known be amenable to both a highly agentive deontic reading or a possibility interpretation (Höhle 1978; Steinbach 2002; Holl 2010). Our item *sehen* ('see') is more amenable to a dispositional possibility reading, yet does not categorically rule out deonticity.

Regardless, an LMM utilizing treatment coding (setting *gehen* and *von* as reference baselines) tested the interaction between VERB CONSTRUCTION and ADJUNCT TYPE. The model confirmed a highly significant main effect of ADJUNCT: across all constructions, implicit *für*-phrases are significantly preferred over explicit *von*-phrases ($\beta = 0.89$, $SE = 0.09$, $z = 9.64$, $p < .001$).

Crucially, the interaction revealed that the "acceptability gap" between *für* and *von* is significantly narrower for *gehen* than for both the canonical middle ($\beta = 1.29$, $SE = 0.19$, $z = 6.94$, $p < .001$) and the *sein* passive ($\beta = 0.93$, $SE = 0.19$, $z = 5.00$, $p < .001$). We attribute this narrower gap in *gehen* to speaker insecurity stemming from a form-meaning mismatch. Analogically, the *gehen*-construction resembles modal *zu*-infinitives (which permit *by*-phrases), exerting a structural pull that slightly softens the prescriptive penalty for *von*-phrases compared to the unambiguous reflexive middle. Semantically, however, the robust preference for *für*-phrases distinguishes *gehen* from canonical passives. Despite its surface similarity to modal *zu*-infinitives, *gehen* structurally patterns with the middle-voice family in its treatment of implicit agents.

### Instrumental adjuncts

The semantic presence of a syntactically inactive implicit Agent in middles generally licenses instrumental adjuncts (Pitteroff 2014). To verify whether *gehen* aligns with this property, we tested a single stimulus containing an instrumental phrase without an explicit Agent (26).

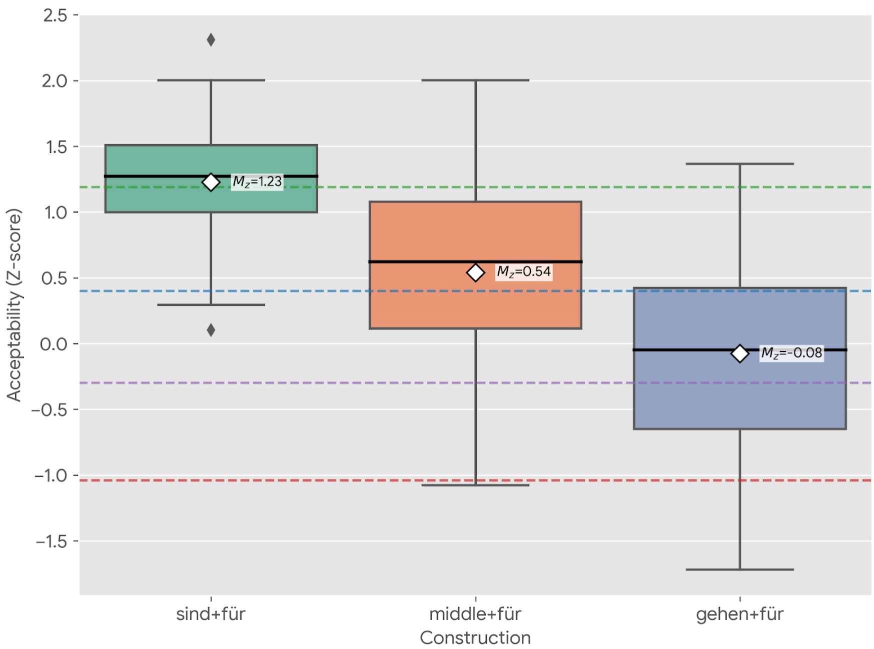



**Figure 5:** Distribution of acceptability ($Z$-scores) across three target constructions comparing explicit agentive *von*-phrases with implicit *für*-phrases

(26) *Die Schraube geht aber mit einem besonderen Schraubenzieher leicht zu lösen.*
the.NOM screw.NOM goes but with a.DAT special.DAT screwdriver easily to loosen
'The screw can easily be loosened with a special screwdriver.'

An instrumental adjunct with [*gehen*+*zu*-INFINITIVE] fell securely within the marginally licensed zone for *gehen* ($M_z = 0.21$). Because only a single item was tested, an intercept-only regression model was utilised to compare the raw ratings against the absolute experimental floor ($M = 1.32$). The model confirmed that the instrumental condition is structurally licensed significantly above the categorical rejection threshold ($\beta = 1.89$, $SE = 0.16$, $t = 11.85$, $p < .001$).

The felicity of the instrumental adjunct indicates that [*gehen*+*zu*-INFINITIVE] licenses an implicit Agent at the semantic interface, consistent with a periphrastic middle analysis. On this account, agent-oriented adverbs should pattern with *von*-phrases in being ungrammatical — a prediction we address qualitatively in Section 4.

## 3.3 Lexical-semantic restrictions

### Adverbial Modification

If *gehen* functions as a middle, it should exhibit preferences for facilitative adverbial modification (Roberts 1987; Condoravdi 1989; Schäfer 2008). To map its licensing profile, we tested a range of adverbial and negative modifiers. The results, visualized in Figure 6 (see Table 8 in Appendix A for full descriptive statistics), confirm that *gehen* requires specific scalar qualification that sets it apart from canonical passive structures.

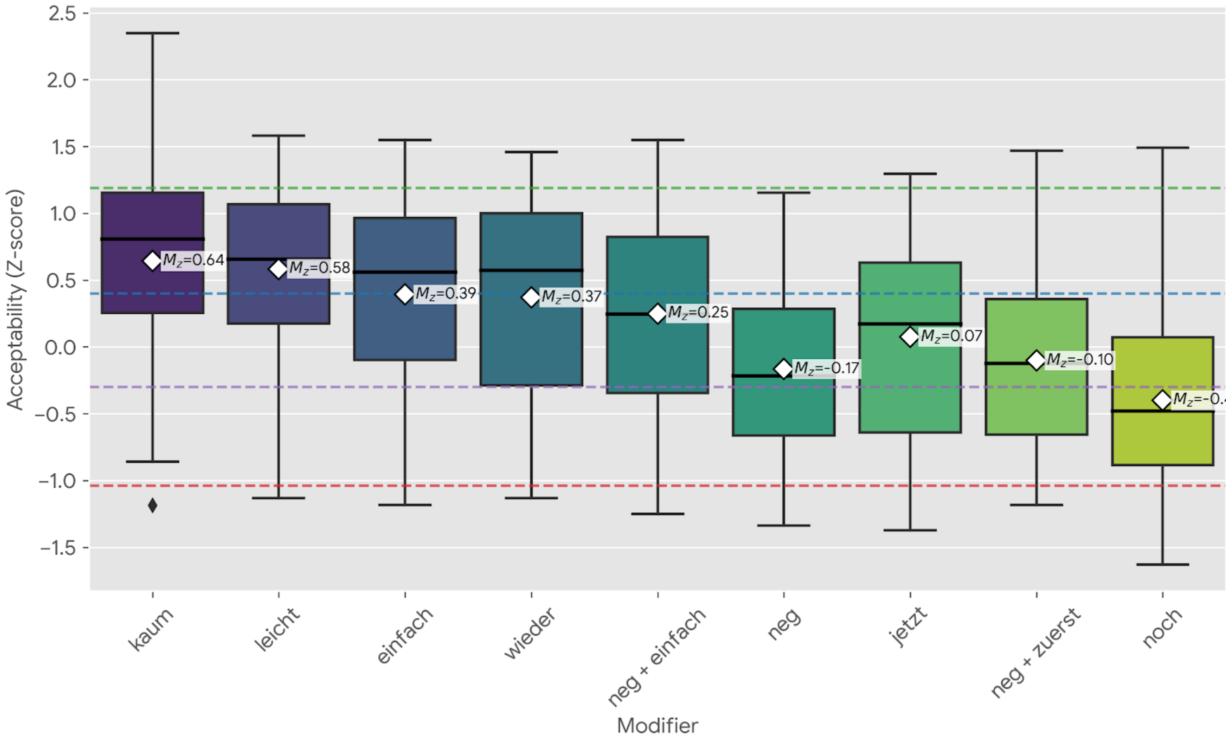



**Figure 6:** Mean acceptability $Z$-scores for adverbial/negative modification of *gehen*

The hierarchy indicates that *gehen* is primarily licensed by modal and degree qualification. Modifiers explicitly linked to scalar potential (*kaum*, *leicht*, *einfach*) elevated the construction securely into the fully licensed zone. Similarly, the relative success of restitutive *wieder* (27) stems from its function of targeting a predicate's result state (Von Stechow 1996; Dobler 2008; Alexiadou and Schäfer 2011); The subject's inherent dispositonal potential as a gradable property is facilitated by presupposing the restoration of a previously held state.[6]

| (27) | *Die* | *Tür* | *geht* | *jetzt* | *wieder* | *auf-zu-schließen.* |
|---|---|---|---|---|---|---|
| | the.NOM.SG.F | door | go.3SG.PRES | now | again | up-to-close.INF |

‘The door can be locked/unlocked again now.’

Crucially, this modal licensing requirement disproves that [*gehen*+*zu*-INFINITIVE] is fundamentally associated with negation as an “impossibility anticausative ” (Cysouw 2023), in addition to showing general behaviour inconsistent with an anticausative analysis. The experimental results demonstrate that bare negation (*gehen_neg*) provides only a marginal improvement ($M_z = 0.14$). In fact, adding a modal adverb to a negated structure (*einfach nicht*,

6. Scores for *wieder* revealed a bimodal distribution, suggesting a divergence between speakers who successfully accessed a restitutive reading and likely those who interpreted it as temporal repetitive marker.

$M_z = 0.25$) measurably boosts acceptability over negation alone. Negation can simply "rescue" an otherwise illicit structure by forcing a modal reading, a behaviour consistent with canonical middles (for middles see Roberts 1987; Condoravdi 1989; Schäfer 2008; Lekakou 2005). The modal and scalar potential structurally anchors the construction.

Conversely, purely temporal framing proved insufficient. The adverbs *jetzt* and *zuerst* landed firmly in the marginal/rejected zones, demonstrating widespread structural failure. The most severe degradation occurred with the modifier *noch* 'still' (28).

(28) *Der Quark geht noch zu essen*
the.NOM quark.NOM go.PRES.3SG still to eat.INF
'The quark is still edible / can still be eaten.'

This degradation sheds light on *gehen*'s lexical-semantic constraints. Although verbs like *essen* 'eat' can be compositionally coerced into telic Accomplishments (Vendler 1957), the root itself is an inherently atelic manner-activity verb (see MANNER/RESULT COMPLEMENTARITY in Rappaport Hovav and Levin 2010). This inherent atelicity triggers a profound mismatch with the modifier *noch*. Unlike evaluative and degree modifiers that actively induce a gradable scale, the scalar focus particle *noch* is too weak to license a scale independently. Instead, it acts as a presupposition trigger requiring a pre-existing scalar or phasal structure enabled by the lexical verb itself (König 1991:52–56). Consequently, when paired with a manner root like *essen*, the presupposition fails because the verb lacks inherent scalar infrastructure. The categoric penalization in the data ($M_z = -0.40$) reflects the cognitive cost of attempting to coerce a scale of "edibility" where the underlying lexical material provides no support. Therefore, *noch* is predicted to be grammatical when supported by inherently resultative roots (e.g., *reparieren* 'repair'). Indeed, we find such examples in natural data (29). This confirms that *gehen* interacts fundamentally with the lexical aspect of its complement, a restriction we unpack in the following section.

(29) *vieles geht noch zu reparieren!* [7]
much.NOM goes still to repair.INF
'Many things can still be repaired!'

To statistically evaluate this hierarchy, an LMM with a planned contrast was utilized (setting temporal/aspectual modifiers as the reference baseline). The model confirmed that modal and degree modifiers generated a massive, highly significant structural advantage ($\beta = 1.04$, *SE* $= 0.09$, $z = 11.20$, $p < .001$). While canonical middles are generally restricted to evaluative manner adverbs, the unique preference of *gehen* for degree adverbs (*kaum*) and restitutive aspect (*wieder*) empirically confirms its specialization as a functional raising head measuring the realisability of a potential state.

7. Source: Nissanboard, https://www.nissanboard.de/forum/index.php?thread/100011-problem-mit-der-hinterachse-an-meinem-gti-hilfe/&postID=2033330. Posted October 24, 2008.

**Vendler's verb classes**

Canonical middles show a preference for Accomplishments, are compatible with Activities, but categorically reject Achievements (Roberts 1987; Fagan 1992; Ackema and Schoorlemmer 1995; Lekakou 2005; Pitteroff 2014). To test whether *gehen* observes these same aspectual restrictions, we analysed acceptability scores across all stimuli representing Accomplishments ($n = 6$), Activities ($n = 3$), and Achievements ($n = 5$). The results confirmed a strong preference for inherent Accomplishments (30a), while Achievements were categorically rejected (30b).

(30) a. *Die Tür geht jetzt leicht aufzuschließen*
The door goes now easy up-to-close.INF
'The door can easily be unlocked now.'

b. **Der Schaffner geht nicht zu überzeugen.*
the conductor go not to convince.INF
Intended: 'The conductor cannot be convinced.'

Because scalar adverbial boosters (e.g., *leicht*, *einfach*) were unevenly distributed across the verb classes in our stimuli, raw averages risk conflating the aspectual capacity of the verb with the independent licensing power of the adverb. An LMM incorporating adverbial type as a fixed-effect covariate was therefore used to partial out this variance and isolate the underlying aspectual pattern.

Even after controlling for the independent contribution of adverbial modification ($\beta = 0.47$, $SE = 0.08$, $z = 5.54$, $p < .001$), the aspectual hierarchy remained intact. Accomplishments ($M_z = 0.32$) significantly outperformed Activities ($M_z = -0.07$; $\beta = 0.45$, $SE = 0.09$, $z = 4.81$, $p < .001$). Both Accomplishments ($\beta = 1.25$, $SE = 0.08$, $z = 16.59$, $p < .001$) and Activities ($\beta = 0.80$, $SE = 0.11$, $z = 7.48$, $p < .001$) showed substantially higher acceptability than Achievements ($M_z = -0.61$).

The aspectual preference persists under adverbial boosting. For example, while the unbounded Activity *spielen* 'to play' is boosted into the licensed zone by *leicht* ($M_z = 0.34$, 31a), it still underperforms compared to an equivalently boosted Accomplishment like *aufzuschließen* 'to unlock' ($M_z = 0.58$, 30b). However, future work must investigate different activities or rather manner roots, more rigorously.

(31) a. *Jimmys Gitarre geht leicht zu spielen.*
Jimmy's guitar goes easy to play.INF
Intended: 'Jimmy's guitar plays easily.'

Achievements resist adverbial boosting entirely. Even when intensified by the evaluative scalar adverb *einfach* ($M_z = -0.69$), they remain overwhelmingly less acceptable than standard, non-intensified Accomplishments ($M_z = 0.14$). This confirms that the lexical-semantic constraint against punctual (and thus ungradable) Achievements overrides the licensing power of adverbs.

Together, these results indicate that *gehen* requires a complement encoding a gradable result state. The punctual structure of Achievements provides no scalar domain for the dispositional operator to evaluate, accounting for their categorical rejection.

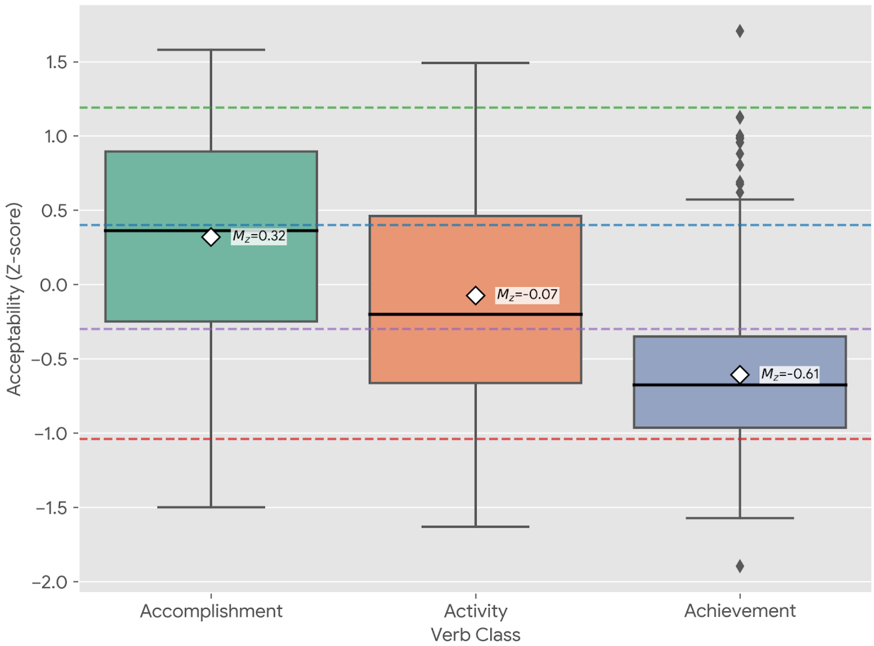



**Figure 7:** Distribution of acceptability (Z-scores) across verb classes under *gehen*

# 4 Discussion

We now address the theoretical implications of the results, organized by domain: clausal syntax, argument structure, and lexical semantics. We ultimately argue that *gehen* operates as a functional raising head that enforces detransitivization through unaccusative syntax.

However, given the experiment's design as a pilot study, the experimental item set was restricted to ensure a manageable task load while maintaining as high statistical power as possible. Thus, where necessary, we explore additional attested restrictions from related (medio-)passive constructions, via post-hoc exploratory observations based on native-speaker judgements.

## 4.1 Clausal Architecture and Selection

Evidence for the functional status of *gehen* stems from its resistance to extraposition. This behaviour aligns with functional raising verbs and modal passives, which form a coherent verb cluster or a single prosodic phrase with their complement. Haider (see 2010:335f) argues these constructions resist extraposition because the matrix verb forms a complex head with the lexical verb. Alternatively, Wurmbrand (2001) proposes they form a single prosodic phrase.

In contrast, LRIs in Long Passives do sometimes permit extraposition (see also Wöllstein-Leisten 2001; Haider 2010; Müller 2020). Assuming a ban on Extraposition to be a diagnostic of functional status, *gehen* aligns with a functional raising head; we note that we understand functional status as reflecting the inability to assign thematic roles, and not a requirement for *gehen* to lexicalize a non-verbal functional projection in T.

(32) *Louisa is very happy that …*
…**dass [das Spiel] endlich geht [zu installieren].*
…that [the game.NOM] finally [goes] [to install].
'…that the game can finally be installed.'

(33) *… dass [der Hund] vergessen wurde, [zu füttern].*
… that the dog.NOM forgotten was, to feed.
'that it was forgotten to feed the dog.' (ex.2c Haider 2010:285)

However, to understand the exact size of this complement, we must introduce a post-hoc qualitative observation regarding auxiliary selection. Auxiliaries in the infinitival domain of *gehen* are strongly rejected (34).

(34) **Das Fahrrad geht (bis morgen) leicht repariert zu sein/haben.*
The bike goes (by tomorrow) easily repaired to be/have.INF
Intended 'The bike can be easily repaired by tomorrow.'

We note that *scheinen* allows auxiliaries after *zu* (35a), while modal *sein+zu* (Haiden 2005; Holl 2010) (35b) and LRIs are incompatible with auxiliaries in the infinitival domain (35c).

(35) a. *Sie scheint das Buch gelesen zu haben.*
She seems the book read.PST.PTCP to have.AUX.INF
'She seems to have read the book.'

b. **Sie ist (bis morgen) fotografiert zu sein / zu haben / worden zu sein.*
she is (by tomorrow) photographed to be / to have / been to be
Intended: 'She must have been to be photographed by tomorrow.'
(ex.30 Holl 2010:49)

c. …**dass schon [der Traktor und der Lastwagen] repariert zu haben vergessen wurden.*
…that already [the tractor and the truck.NOM] repaired to have.INF forgotten were.PASS.3SG.PL
'that it was forgotten to have already repaired the tractor and the truck.'
(ex.240b Wurmbrand 2001:300)

The difference comes from the ability of *scheinen* to select a TP complement, while LRIs and *sein+zu* can only take a truncated ("voiceless") vP complement (see Wurmbrand 2001:3,79).

This restriction against auxiliaries in the embedded domain, shared by LRIs and *sein+zu*, indicates a selectional restriction on *gehen* for a vP complement without functional structure (see Wurmbrand 2001; Holl 2010; Haider 2010).

Notably, *gehen* also restricts its upper domain—specifically, as matrix compound tense morphology is ungrammatical. This restriction is shared by the raising verb *scheinen*. Wurmbrand (2001) considers *scheinen* as an epistemic modal operator occupying a position above T and argues that the auxiliary cannot scope over *scheinen*, thus ruling out compound tense. From a hierarchical perspective, the compound tense prohibition with *gehen* cannot be derived from a high-operator status. If *gehen* is a modal introducing a disposition, it qualifies as a dynamic possibility operator. Dynamic modals sit at the upper edge of the verbal complex under T (Wurmbrand 2001; Cinque 2006b) and are therefore compatible with compound tense. Consequently, a different mechanism is necessary by which *gehen* resists temporal auxiliaries. We return to this mechanism in Section 5.

## 4.2 Lexical Semantics

In lexical semantic terms, the acceptability of [*gehen+zu*-INFINITIVE] is conditioned on two intersecting requirements: i) modal qualification in the form of a need for scalar potential, and ii) aspectual compatibility via a selective requirement for a lexical root inherently encoding a result state.

First, our analysis of adverbial modification allows us to convincingly reject the construction's classification as an "impossibility anticausative" (Cysouw 2023). If the construction were truly defined by 'impossibility,' negation should logically outperform modal degree modifiers. Rather, the substantial advantage of *kaum* over pure negation ($\beta = 0.50, z = 5.48, p < .001$) demonstrates that negation is merely tolerated as a baseline. This strengthens the case that *gehen* is inherently linked to scalar potential rather than absolute impossibility.

Second, we found that this requirement for scalar potential directly dictates the construction's lexical-semantic constraints. By controlling for the independent licensing power of adverbs via a Linear Mixed-Effects Model, we observed a rigid aspectual hierarchy aligning with Vendler's (1957) verb classes. Lexical Accomplishments possessing an inherent result state constitute the most productive predicates in the construction. The established dispositional quality of the construction, in line with its unequivocal preference for Accomplishments strongly supports its classification as a medio-passive strategy.

Crucially, the dispositional modal *gehen* requires a gradable scale. This explains the strict rejection of Achievements. Because they denote strictly punctual events, they are inherently non-gradable (Vendler 1957, 1967); they encode a non-gradable two-point scale (Beavers 2008; Rappaport Hovav 2008). Therefore, the punctual change denoted by an Achievement cannot be mapped onto the durative scalar potential that the dispositional modal *gehen* strictly requires.

Similarly, this scalar requirement explains the marginality of Activity verbs and compositional Accomplishments. Manner roots, which are inherently Activities, are only marginally tolerated under appropriate adverbial modification (e.g., with *leicht*), because their atelic nature can occasionally be coerced into a bound, gradable reading. We saw that *noch* was too weak to force this coercion, even when combined with a lexical Activity verb like *essen* that can receive an Accomplishment reading compositionally with a quantized object, e.g., der/einen Apfel 'the/a apple'; telicity in such contexts is calculated semantically via the DP measuring out the event (see Tenny 1994), yet our data reveal a selectional difference between verbs that denote Accomplishments via default and activity verbs which may compositionally come Accomplishments. We propose a mechanism to account for this difference in the next subsection.

Having established these rigid lexical-semantic parameters, the theoretical challenge is determining how the grammar actually enforces them. As we will see, these aspectual restrictions cannot be reduced to simple C-selectional or S-selection properties. We must therefore turn to the syntactic architecture of the construction to understand how the grammar structurally accommodates these roots and blocks incompatible event configurations.

## 4.3 Argument structure: Deriving syntactic and semantic restrictions

We now turn to an exploration of the argument-structural properties of [*gehen*+*zu*-INFINITIVE]. This section sets out the syntactic nature of infinitival verbal complex under *gehen* and *gehen*'s selectional properties. This discussion of the argument-structural properties supports the steadfast classification of [*gehen*+*zu*-INFINITIVE] as a periphrastic middle strategy. Moreover, we develop a configurational account that derives lexical-semantic and syntactic restrictions in the complement domain.

#### Properties

As established in the experimental data, the *von*/*für* asymmetry unequivocally demonstrates that *gehen* does not project a syntactic AGENT argument. The categorical rejection of overt agentive *von*-phrases aligns *gehen* strictly with canonical reflexive middles (Steinbach 2002; Lekakou 2005; Schäfer 2008), sharply distinguishing it from both *werden*-passives and *lassen*-middles, which freely license explicit agents. Likewise, we showed that *sein*+*zu* passives are compatible with *von*-phrases, yet the acceptability may be affected by the modal interpretation and perhaps the arbitrariness of the agentive adjunct. However, the comparative acceptability of *für*-phrases in [*gehen*+*zu*-INFINITIVE] strongly suggests that an implicit Agent is still present semantically or can be reconstructed at the Conceptual-Intentional (C/I) interface, even if it is completely absent from the syntax, as is the case for Middles (see Fagan 1992; Ackema and Schoorlemmer 1995, 2015; Steinbach 2002; Lekakou 2005; Pitteroff 2014; Schäfer 2008). This position was further supported by the confirmed felicity of an instrumental adverbial adjunct, which is likewise dependent only on the presence of a semantic Agent.

To corroborate the absence of a syntactically active agent, we conducted post-hoc qualitative explorations using classic syntactic diagnostics for the nature of arguments and the presence of a syntactic Agent (Roberts 1987; Roeper 1987; Baker et al. 1989; Bhatt and Pancheva 2006; Alexiadou et al. 2015). Unsurprisingly, *gehen* strictly prohibits both modification by agent-oriented adverbs (e.g., *sorgfältig* 'carefully') and control into *um ...zu* purpose clauses (36).

(36) a. **Das Spiel geht sorgfältig zu installieren.*
The game goes carefully to install
Intended: 'The game can be carefully installed.'

b. **Das Spiel geht einfach zu installieren, um Zeit zu sparen.*
The game goes simply to install in order time to
Intended: 'The game can be installed simply, in order to save time.'

If an Agent were structurally present, the sentences in (36) should be grammatical. Their failure suggests that the restructured infinitival vP domain under *gehen* and the verbal complex housing *gehen* itself are both genuinely voiceless, i.e., lacking a projection capable of hosting a silent syntactic Agent.

Furthermore, as confirmed by the data, argument-structural restrictions on internal arguments familiar from canonical middles extend to *gehen*, though the *gehen*-configuration proves to be even more structurally constrained. *Gehen* categorically resists dative-assigning predicates, including ditransitives. For monotransitive dative assigners, we already identified an aspectual confound in Section 3 concerning the Activity status of *helfen* ('to help'). Although a restriction against dative Case or a second internal argument is a well-documented hallmark of German(ic) canonical middles (see Ackema and Schoorlemmer 1994, 1995, 2015), German middles can exceptionally preserve dative Case with dative-assigning verbs in impersonal configurations. However, *gehen* categorically rejects dative arguments in their entirety. This absolute structural ban contrasts with modal [*sein*+*zu*-INFINITIVE], which freely permits dative preservation, indicating that the *gehen*-configuration is architecturally distinct and highly restricted in its internal selectional capabilities.

The incompatibility of ditransitives shall prove more informative for a structural analysis. Firstly, double-object configurations are well-known to be strictly forbidden in canonical middles (Roberts 1987; Fagan 1992; Ackema and Schoorlemmer 1994, 1995, 2015); within the Government and Binding framework, this restriction was standardly derived via Case-theoretic mechanisms (see Roberts 1987) or stipulated as an Affectedness Constraint (see Fagan 1992). An exact translation of these specific middle-voice constraints into contemporary terminology is unnecessary here, as we argue that a distinct, strictly configurational motivation is at play for *gehen*.

**Restrictions on Selection**

Recall the analyses by Wurmbrand (2001, 2015) that LRIs in long passives select an agentless verbal complex, i.e., a specifierless VoiceP (bare VP in earlier work). At first sight, we might simply extend this condition to *gehen*. However, that would be too simplistic. Indeed, a ditransitive can appear in the infinitival domain of a stripped-down verbal complex in a LRI, as shown by scrambling data from extraposed infinitives in (see 37) (see Wurmbrand 1998:116); for example, the accusative Theme of ditransitive *übermitteln* 'to send/transmit' can scramble into the matrix clause, stranding the dative goal within the truncated embedded domain.

(37) *weil der Hans [einen Brief]$_{SCR}$ versucht hat [der Maria $t_{SCR}$ zu übermitteln]*
since the John [a letter].ACC tried has [to Mary.DAT to send]
'since John tried to send a letter to Mary' (ex. 18 Wurmbrand 1998:116)

Indeed, following Wurmbrand's (2015) updated theory, which does not rely on truncation (contra Wurmbrand 1998, 2001), but a specifierless expletive VoiceP, both ditransitives with low and high applicatives under VoiceP (Pylkkänen 2008) are predicted to be permissible in such contexts. Indeed, consultants accept an equivalent LRI containing scrambling, extraposition and a high applicative (38), although the convoluted nature of scrambled extraposed LRIs degrades all such structures somewhat.

(38) *weil der Hans [ein Haus]$_{SCR}$ versucht hat [seinen Eltern $t_{SCR}$ zu bauen]*
since the John [a house].ACC tried has [his parents].DAT tSCR to build
'since John tried to build a house for his parents'

Shifting back to [*gehen*+*zu*-INFINITIVE], the empirical diagnostics clearly demonstrate that *gehen* cannot select an agentive VoiceP. However, the absence of Voice does not structurally exclude applicatives. If we stipulate a condition that [*gehen*+*zu*-INFINITIVE] can select only a vP,[8] we rule out high applicatives but not low applicatives. Recall the rejected ditransitive example in (39).

(39) a. **Die Email geht ihm nicht zu schicken.*
the email goes him.DAT not to send
Intended 'The email cant be sent to him.'

The ditransitive *schicken* 'to send' is both a low applicative (Pylkkänen 2008) and an Accomplishment; hence, it cannot fail for lexical-semantic reasons. Thus, a vP-only selection alone does not exclude low applicatives. However, as shown in (40), low-applicatives are incompatible with resultatives, Pylkkänen (2008:40ff), because low-applicative arguments cannot establish a relation to the subject of a small clause, i.e., a ResultP.

8. We treat vP as simply the introducer of the event and the verbaliser (Kratzer 1996; Embick 2004a,b)

(40) a. He painted me this flower. Low Applicative

b. He painted this flower blue. Resultative

c. *He painted me this flower blue *Combination

(ex.69 Pylkkänen 2008:40-1)

Therefore, we can posit the following restriction on *gehen*:

(41) **A selectional restriction on *gehen***:
The functional raising head *gehen* selects only for a resultative verbal complex involving a voiceless vP complement and ResultP.

We must ask how a functional raising head can impose such strict aspectual requirements on its vP complement. We might stipulate that *gehen* C-selects an inherently resultative $\text{v}_{\text{CAUSE}}$ which necessarily selects a ResultP small clause (cf. von Stechow 1995; Folli and Harley 2005, 2007; Alexiadou et al. 2006b,a). However, such a C-selectional rule cannot force a restriction for Accomplishments, while excluding Achievements. Moreover, a standard 'flavors of $v$' analysis (see e.g., Folli and Harley 2005) considers $v$-cause to be inherently agentive, thus requiring us to stipulate a special defective unaccusative $v_{\text{CAUSE}}$ (but see Pylkkänen 2008). Therefore, we adopt the updated configurational approach by Alexiadou et al. (2015) in which causative semantics are not defined in the narrow-syntactic structural features of a head. Instead, causation is a post-syntactic interpretation derived at the Conceptual-Intentional (C/I) interface whenever the syntax forms a "telic pair" $<e_1, e_2>$ (see Higginbotham 2000).

### Beyond Selection

How then can we derive the documented lexical-semantic and argument-structural restrictions on [*gehen*+*zu*-INFINITIVE] without recourse to C- or S-selectional restrictions that span across multiple projections? In the modular architecture of Distributed Morphology (Halle and Marantz 1993) employed by Alexiadou et al. (2015), Narrow Syntax is blind to the encyclopedic content, i.e., meaning of roots. Crucially, roots themselves do not possess inherent aspectual classes such as Accomplishments or Activities; rather, these classifications arise configurationally. Because the computational system cannot look ahead or select roots based on their encyclopedic content, it blindly derives functional structures. The resulting syntactic complex is then sent to C/I for interpretation. If the syntax fails to derive the appropriate bieventive configuration (a $v$P dominating a ResultP), or the encyclopedic content of the inserted root is conceptually incompatible with the dispositional requirements of *gehen*, the derivation is illegible and simply crashes at the interface. Let us then turn to the specific structural mechanics of this derivation.

We can safely assume that *gehen* C-selects an eventive vP, but strictly forbids the projection of an agentive Voice layer (see Kratzer 1996; Alexiadou et al. 2015). Yet, to satisfy the Accomplishment requirement, the derivation must generate an eventive $v$ head that merges directly with a ResultP. Crucially, *gehen* must blindly select this vP/ResultP configuration, or rather it must rule out any other configuration.

The aspectual and conceptual restrictions, we argue, are exclusively enforced as interface conditions at LF and C/I. First, LF requires the *v*-to-ResultP configuration to compute the necessary transition associated with a gradable scale. Second, the semantics of modal *gehen* require both a conceptual external force (an implicit agent) to establish the dispositional relation to the promoted Theme, and a conceptually measurable process to evaluate scalar potential (cf. Condoravdi 1989; Tenny 1994; Beavers 2008). That is, a disposition is inherently relational as a Theme can only be evaluated as 'facilitating' or 'impeding' an event if there is a potential actor (cf. Ackema and Schoorlemmer 1994), i.e., implicit Agent to realise that event.

Conversely, Activities and compositional Accomplishments fail because their eventive v does not project a ResultP, depriving LF of the configuration required to anchor a change of state. This explains why compositional Accomplishments coerced from inherent Activities and thus manner roots are rejected with *gehen*, e.g., *einen Apfel essen* ('to eat an apple') (Vendler 1957; Tenny 1994). In a compositional Accomplishment, telicity is derived via an event-bounding functional projection, InnerAspP, situated below v but above the root (MacDonald 2008; Travis 2010) (see 42).[9] While this aspectual head successfully calculates the quantitative measuring-out of the event, it merely provides a telic boundary; crucially, it does not project a ResultP denoting a distinct target state (*s*). Consequently, when modal *gehen* probes its complement for a state to evaluate, it encounters the event boundary at InnerAspP but fails to find the required *s* variable. Consequently, a dispositional relation cannot be established. Thus, because the narrow syntax fails to provide the required result state, the derivation crashes at LF regardless of the quantized object's telic boundary.

(42) [vP [InnerAspP [ VP ...]]]

However, we must comment on apparent exceptions to the rule involving the atelic root *spiel* ('play') in (43).

(43) a. *Jimmys Gitarre geht leicht zu spielen.*
Jimmy's guitar goes easy to play.INF
Intended: 'Jimmy's guitar plays easily.'

The acceptability of seemingly atelic roots like $\sqrt{\text{SPIEL-}}$ ('play') in (43) does not violate the structural restrictions of *gehen*; rather, it provides clear evidence for structural coercion and highlights the division of labour between the narrow syntax and the semantics. Recall that verbs of consumption (e.g., 'to eat an apple') involve an incremental Theme. This forces the syntax to project an InnerAspP in order to measure out the event, which subsequently starves the modal of a required target state. In contrast, an instrument like a guitar is not an incremental theme; its physical volume does not measure out the event. Consequently, the syntax is not forced to build an aspectual measuring node. Since narrow syntax freely generates

9. See also Borer (2005) for a similar but crucially exoskeletal approach.

structures, it can merge the acategorial root into a resultative configuration, e.g., with a ResultP. We have established that although *gehen* C selects an eventive vP complement, it is blind to anything below this. The crucial filtering occurs at C/I, when *gehen* evaluates the vP. Here, it successfully locates the target state encoded by ResultP. This conceptual coercion succeeds because the guitar is designed to be physically manipulated from an inert and silent state into an active and loud state. The dispositional evaluation is thus uninhibited and the derivation survives. Ultimately, this proposal predicts that typically atelic roots remain compatible with *gehen* provided that no incremental Theme and its associated *InnerAspP* are present, thereby permitting coercion into a resultative structure.

Furthermore, following the spirit of Alexiadou et al. (2015), the roots of verbs successfully licensed by *gehen* can be encyclopedically classified as either externally caused roots (e.g., $\sqrt{\text{REPAIR}}$, $\sqrt{\text{INSTALL}}$) or cause-unspecified roots that permit a durative process (e.g., $\sqrt{\text{OPEN}}$.[10] Both classes are conceptually compatible with an entailed implicit Agent at the C/I interface, a key ingredient for middle formation (see Condoravdi 1989; Schäfer 2008). If the syntax merges an internally caused or purely spontaneous root (e.g., $\sqrt{\text{ROT}}$) inside the vP, the derivation survives the syntax but triggers a semantic mismatch at C/I; internally caused roots are conceptually at odds with external control, bleeding entailment of an implicit Agent. Similarly, if the syntax merges a punctual Achievement root (e.g., $\sqrt{\text{FIND}}$, $\sqrt{\text{BREAK}}$), it provides the necessary change of state but lacks an encyclopedically durative process, leaving the dispositional modal without a time span to evaluate; consequently, gradable scalar potential is ruled out.

Furthermore, recall that *gehen* is neither dominated by nor selects a VoiceP; the absence of Voice means there can be no syntactically active implicit agent, which explains why *von*-phrases, agent-oriented adverbs, and control into purpose clauses are all ungrammatical. Finally, as discussed, if the syntax attempts to build a low applicative in the infinitival domain, the resulting configuration structurally blocks the formation of the required ResultP (Pylkkänen 2008:40-41). Overall, a clear picture emerges in which *gehen* acts as a middle raising head that forces detransitivization via unaccusative syntax (see also Abraham 2020); it merges as an unaccusative head without VoiceP above it and selects a bare vP/ResultP configuration. In short, both levels lack the functional Voice architecture associated with canonical passives and active causatives (Alexiadou et al. 2015).

Let us now turn to set out a syntactic analysis of the entire derivation.

# 5 Analysis: From CP to ResultP

It is now beyond doubt that [*gehen*+*zu*-INFINITIVE] is a highly restrictive infinitival middle. Likewise, in Section 4.3, we already proposed a theoretical framework with which to analyse the

10. For foundational discussions on the conceptual ontology of roots and the distinction between externally caused, internally caused, and cause-unspecified eventualities, see Levin and Rappaport Hovav (1995) and Levin and Rappaport Hovav (2005)

lexical vP domain under *gehen* and derive argument-structural and lexical semantic restrictions. We now present a step-by-step derivation, using (44) as an example derivation.

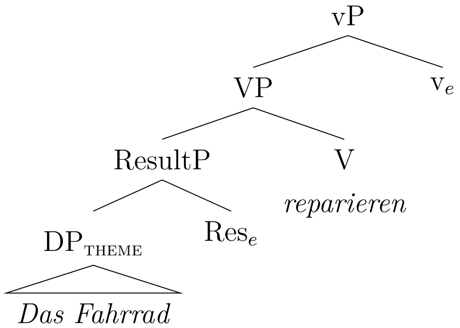
(44) *…dass Das Fahrrad leicht zu reparieren geht.*
…that the bike.NOM easily to repair goes
'…That The bike is easy to repair.'

## 5.1 the vP domain: a recap

We have established that [*gehen*+*zu*-INFINITIVE] fails all tests of syntactic agentivity. We thus posit that *gehen* behaves like a functional raising verb which selects an eventive vP, forcing an unaccusative structure. The infinitival vP merges without a dominating agentive voice layer and thus represents a truncation of an otherwise canonically agentive causative predicate (*pace* Wurmbrand 2001).

We show the derivation of the infinitival complex in (44). We use the label $\text{v}_e$ to denote eventive status (cf. Alexiadou et al. 2015). We do not employ V-to-v movement, as it would be a linearly superfluous and dubious operation in German (Haider 1993, 2010; Broekhuis 2023). Moreover, recall the restriction for compatible Accomplishments to appear in inherently resultative configurations, e.g., change-of-state verbs such as *reparieren* 'to repair', as opposed to compositional Accomplishments, e.g., *essen* 'to eat' which comprise manner root with a quantised incremental Theme. We argue for a restriction for bieventive results that lexicalize a gradable scale. Descriptively, this aligns with a well-attested requirement for causatives to take a small-clause ResultP complement (cf. Hale and Keyser 1993; Von Stechow 1996; Hale and Keyser 2002; Folli and Harley 2005; Harley 2005; Alexiadou et al. 2006b). ResultP, we argue, is headed by $\text{Res}^0$ which introduces the secondary event variable $<e_2>$, structurally deriving the telic pair $<e_1, e_2>$, cf. Higginbotham (2000). This is necessary because the DP is an entity and not an event. Likewise, following standard event-structural analyses of Germanic separable particle verbs (den Dikken 2007; McIntyre 2004; Kratzer 2005), we analyse $\text{Res}^0$ as the locus for the realization of the particle (cf. Ramchand 2008).

(45)



## 5.2 Adding *zu*

One question is how *zu* enters the derivation and what its role is. In our analysis *zu* does not perform or license any functional or semantic operation. Our principle claim is that the entirely unaccusative syntax of the truncated vP provides a passive interpretation for free (see Abraham 2020). This claim differentiates [*gehen*+*zu*-INFINITIVE] from [*sein*+*zu*-INFINITIVE], the latter of which has been argued to employ *zu* as the passivizer. For Holl (2010) *zu* is a passivizing head, while for Haiden (2005) and (Haider 2010) it is a passivizing affix which absorbs/saturates the external θ-role. We do not adopt these analyses, since i) they are beyond the scope of [*gehen*+*zu*-INFINITIVE], and ii) it is not clear how a passivising *zu* is compatible within the adopted Voice typology. Moreover, treating *zu* as a passivizing element, be it as an operator or argument-absorber, introduces unnecessary architecture into our derivation. [*gehen*+*zu*-INFINITIVE] lacks any synactic Agent to be absorbed, bound, or saturated. Hence, even if *zu* encodes this function, it is redundant. Our view thus aligns with Wurmbrand (2001) who considers *zu* to lack semantic content but raises the problem of how it enters the derivation, given that it should be ruled out in the numeration, on account of Chomsky's Principle of Full Interpretation, Chomsky (1995:24ff), and prohibited from entering it during the derivation (i.e., the Inclusiveness Condition, Chomsky 1995). We thus illustrate an assumed structure for the *zu* + V construct in (46).

(46)

$V^0$

Inf $V^0$

*zu* *reparieren*

If correct, we must then account for the insertion of a formally inactive *zu*. We propose a PF-repair mechanism, assuming that vacuous items do not enter the derivation in the narrow syntax. Following Adger (2003) and Bjorkman (2011), v head enters the derivation with an unvalued inflectional feature, [uInfl]. When the matrix T head values [uInfl] on a raising verb in the matrix clause, e.g., *gehen*, the embedded v head is isolated (see Wurmbrand 2001) and its [uInfl] remains unvalued. In order to avoid a crash at PF, the morphology must deploy a repair strategy. Following assumptions from Distributed Morphology (see Embick and Noyer 2001; Embick 2010, 2015), the system satisfies this morphological elsewhere condition (cf. Kiparsky 1973) by inserting *zu* as a dissociated morpheme prefixed to v. Thus, the Inf node in (46) is the result of this structural repair at PF, i.e., it is "sprouted" post-syntactically (Halle and Marantz 1993; Halle 1997). However, in a single extended projection containing a finite auxiliary or modal, the valued [Infl] feature on the functional head will be able to value the [uInfl] a lexical v, thus avoiding *zu*. We note a conceptual similarity to the PF-repair approach by Wurmbrand (2015), in which *zu* is the default reflex of a dedicated yet deficient restructuring v. Our approach avoids this added architecture. Therefore, we join approaches that *zu* is prefix on the infinitive (Demske-Neumann 1994,Haider 2010:349-350,Abraham 2020,Haiden 2005)

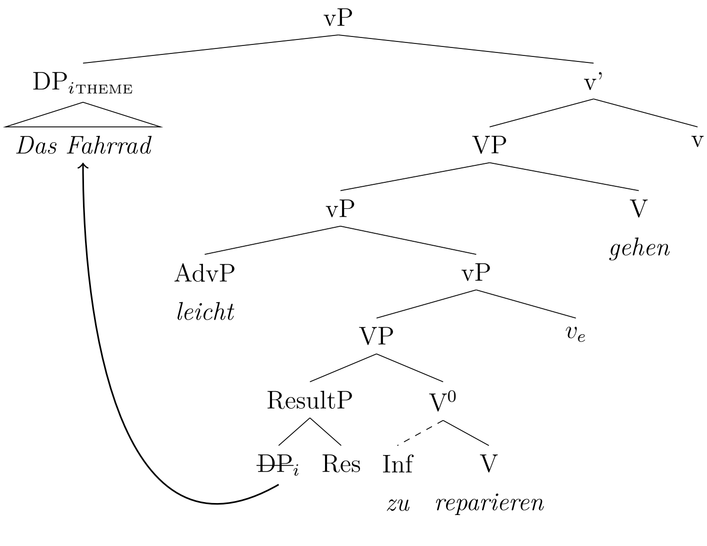
## 5.3 Adding *gehen*

Now *gehen* must combine with the lexical predicate. As established *gehen* behaves like a functional restructuring verb, lacking theta-assigning capabilities and inducing Raising. However, it is more restrictive than *scheinen* 'seem' regarding the nature of the selected complement. This property means that *gehen* appears to retain some lexical content like a semi-functional head, yet in fact its only selected restriction is for an eventive vP. We argued that the lexical-semantic restrictions are derived by the configurational properties of the predicate and its compatibility with *gehen* in deriving a gradable dispositional property assigned to the Theme. In contrast, *scheinen* does not place argument-structural or lexical-semanic constraints on its predicate, even selecting TPs.

Moreover, we treat functional raising verbs as unaccusative v heads (see Section 2.2) (*inter alios* Grewendorf 1989; Chomsky 2001; Davies and Dubinsky 2004; Haider 2010) and not a higher functional projection (*contra* Cinque 2006a; Wurmbrand 2001). This position is supported by our repair account of *zu*; if a raising verb occupied a higher projection, nothing prevents it from valuing [uInfl] on the lexical v, thus bleeding insertion of *zu* at PF. The puzzle is then why *gehen* and *scheinen* resist compound tense, and not why they force restructuring.

The derivation for the entire raising verbal complex is given below (47).

(47)




We argue that the Theme moves to a Spec,vP position, since the standard analysis (based on English) dictates that unaccusative raising verbs are defective, non-phasal, and lack a specifier position (Chomsky 2001). In English, which has a subject-related EPP on T, T can then probe the embedded DP directly as probing is not blocked by the phase edge. This triggers subject raising to Spec,TP. However, German is not standardly assumed to have a subject EPP requirement on TP (Abraham 1993; Haider 1993, 1997, 2010; Sternefeld 2006; Biberauer and

Roberts 2010).[11] Indeed for *scheinen*, there does not appear to be any issue as in-situ subjects are known to be unproblematic.

However, for [*gehen*+*zu*-INFINITIVE] there is good evidence that the DP Theme must raise to a position in the middle field; this is evident in the contrast in (48a,b) which shows [*zu*-INFINITIVE] can only be fronted without the DP theme, which must evacuate to middle field. Moreover, (49) strengthens this conclusion, demonstrating that the Theme must evacuate to a position higher than the evaluative adverb, which we consider a vP level modifier, but lower than CP. We assume this position to be spec,vP, but admit that these data are also consistent with a Spec,TP analysis which we reject on wider theoretical grounds for German.

(48) a. *Zu öffnen gehen die Fenster leicht.*
to open goes the window easily
'The windows are easy to open.'

b. **Die Fenster zu öffnen gehen leicht.*
the windows to open goes easily

(49) *Damals ging (??/*schwer) ein Auto schwer zu reparieren.*
back_then went (difficult) a car difficult to repair
'Back then, a car was difficult to repair.'

We propose that the necessity for displacement is tied to the head as a *gehen*-middle is not a canonical raising construction; it is a dispositional middle.[12] Following proposals for category-flexible structure-building EPP features, e.g., [•X•] by Müller (2014), we argue that this v specifically bears a structure-building [•D•] feature (see also Sluckin 2021). The mandatory [•D•] feature on the raising v forces the Theme into the local predicational configuration with *gehen*, which is in necessary to derive the dynamic dispositional property of the Theme. This analysis explicitly requires us to treat $v_{gehen}$ as a phase head (Chomsky 2001, 2008), adopting v to always be a phase (Arad 2003; Marantz 2001, 2007; den Dikken 2007; Legate 2003, 2014). Notably, treating *gehen* as a phase head lends explanatory strength to our account of *zu* insertion. Namely, once the Theme lands in Spec,vP, the vP phase is complete and $v_{gehen}$ and its complement the embedded vP will undergo transfer to the interfaces (Chomsky 2001, 2008). Subsequently, when the embedded v head with unvalued [uInfl] is sent to PF, the repair strategy is triggered that inserts the dissociated morpheme *zu*.

## 5.4 The functional domain: bleeding auxiliaries

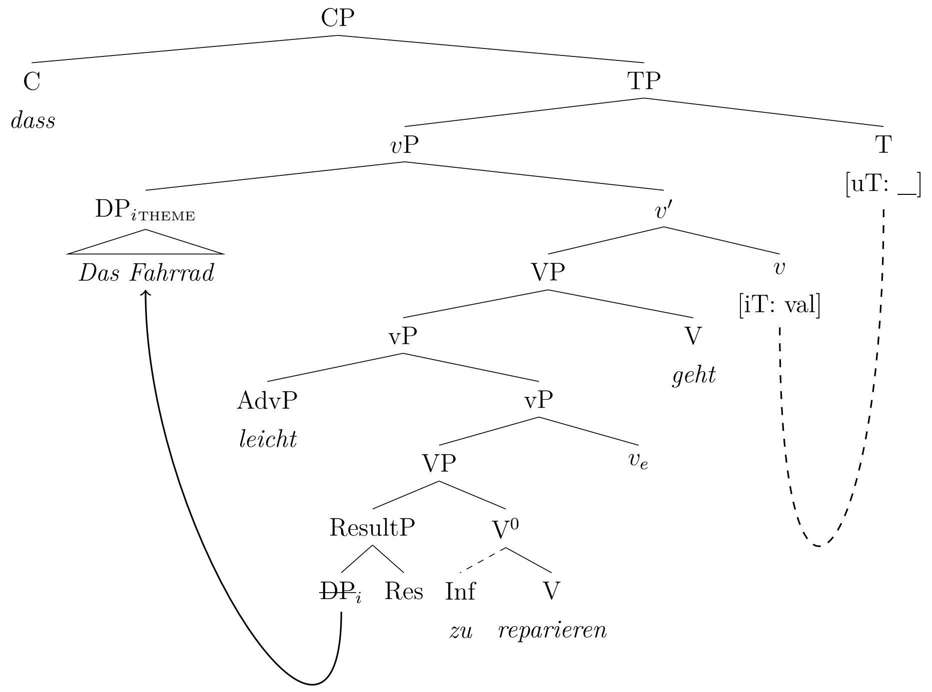
The functional domain, specifically the relationship between matrix T and the raising v, requires separate treatment. Recall that both *gehen* and *scheinen* totally resist compound tenses such

11. This does not mean that Spec,TP never projects, but that probing from T cannot be relied on as the mechanism for subject raising; it is undesirable to reduce Raising to a scrambling-only strategy.
12. We refer to the discussion of micro and nanoparameters by Biberauer and Roberts (2017).

as the perfect. Therefore, we propose a system by which combination of *gehen* with compound past tense morphology is ruled out at the syntax-morphology interface.

To account for the strict incompatibility of [*gehen*+*zu*-INFINITIVE] with compound tenses, we adopt a feature-valuation based approach (cf. Chomsky 2001), assuming that functional raising verbs of this type enter the derivation fully specified for tense (cf. Wurmbrand 2001). Specifically, the functional head *gehen* is merged bearing the valued interpretable feature [iT]. The higher T head enters the derivation bearing unvalued, uninterpretable features [uT:_]. In a standard probe-goal configuration, T probes its c-command domain and locates the matrix $v_{\text{RAISING}}$ head hosting *gehen* (Chomsky 2000, 2001). The unvalued features on T automatically enter into an *Agree* relation with this locally c-commanded goal, receiving their values directly from the [iT] features on *gehen*. Crucially, because this long-distance *Agree* relation successfully values T's features in-situ, it bleeds the generation of a higher temporal auxiliary. Following Bjorkman (2011), temporal auxiliaries operate as default or 'overflow' elements, inserted as a last-resort strategy when the extended verbal projection cannot realize inflectional features. As a result of the raising head valuing T, no features remain that could license or trigger Merge of an auxiliary. The derivation of a full clause is given below (50).

(50)




## 5.5 A final word on modality

Finally, for *sein*+*zu* or *haben*+*zu* modal infinitives, Holl (2010) and Abraham (2020) propose that the modal content is derived from *zu* itself; Abraham (2020) derives modality configurationally from the directional preposition *zu*, while Holl (2010) proposes a homophonous modal-passive *zu*. In contrast, we have already proposed that *zu*, at least for [*gehen*+*zu*-INFINITIVE], is in fact the result of a repair strategy at PF. We have followed the position that *gehen* is a v head encoding dynamic possibility (see also Holl 2010). Holl (2010:10) argues this based on the availability of a possibility reading in simple matrix clauses without a *zu*-infinitive (51).

(51) *Eine solche Übersetzung geht in zwanzig Minuten.*
a such translation goes in twenty minutes
'Such a translation is possible in twenty minutes.'
(ex.5d Holl 2010:10)

However, our data also clearly show that modification via an evaluative adverb, a gradable degree adverb, or negation is required for grammaticality. These two conclusions contradict each other; if *gehen* were the pure source of dynamic possibility, a dispositional reading should be available without further modification.

Let us then consider the feature specification of *gehen*. We assume that *gehen* enters the syntactic derivation bearing a valued, interpretable modal feature [iMod]. At C/I, this feature is interpreted as conveying existential modal force ($\Diamond$) evaluated against a circumstantial modal base which is anchored to the physical properties of the internal argument (see Kratzer 1991; Lekakou 2005; Holl 2010; Pitteroff 2014; Abraham 2020). Since *gehen* selects a bare vP complement lacking a syntactically active agent, an interpretation relying on a deontic ordering source, which requires an entity subject to laws or obligations, is excluded *a priori* (cf. Bhatt 2006). Instead, the pragmatic context provides a stereotypical ordering source that represents the normal progression of events based on the physical properties of the Theme. However, although *gehen* comes preloaded with this modality, it is too weak to yield the necessary dynamic dispositional reading on its own. In order to elicit the dispositional reading, the modality must evaluate the promoted Theme's potential to facilitate the event. This requires a gradable scale to measure exactly how easily that change of state can be achieved. Consequently, the addition of an appropriate adverbial modifier provides the scalar infrastructure needed for the dispositional relation present in dynamic modality, explicitly anchoring this evaluation to the properties of the Theme.

Crucially, our proposals correctly predict the non-dispositional behaviour of *gehen* in control structures. As shown in (52), the dynamic dispositional reading vanishes when *gehen* takes a clausal subject, whether extraposed as in (52a) or fronted as in (52b).

(52) a. *Es geht nicht,* [PRO$_{arb}$ *den Wagen zu reparieren*].
it goes not the.ACC car to repair
'It is not possible/not permitted to repair the car.'

b. [PRO$_{arb}$ *einen Wagen kaputt zu machen*] *geht gar nicht.*
a car broken to make goes at-all not
'It is absolutely unacceptable to break a car.'

In both configurations, the internal argument remains trapped within the embedded Case domain. Because the Theme does not undergo A-movement into the matrix clause, it cannot establish a dispositional relation with *gehen*. Instead, these clausal subjects project a PRO$_{\text{arb}}$ in the active thematic VoiceP dominating the infinitival vP. This structurally provides a higher entity capable of bearing obligation or permission. The existential operator is evaluated against a circumstantial modal base, while the pragmatic context determines the flavour of its ordering source. For example, when evaluated against a deontic (normative) ordering source, it yields a strong social prohibition reading in (52b). Although fundamentally different at the structural level, this proposal resembles then the logic of Holl's (2010) Kratzerian argumentation for variation in the modal interpretation of *zu* in [*sein*+*zu*-INFINITIVE], yet we derive the variation from a combination of the preloaded modal content of *gehen*, the relevant pragmatic context, and the nature of the operation, e.g., Raising vs Control, which is ultimately dictated by the argument structure of the embedded predicate and its thematic properties.

Finally, the arbitrary/generic flavour of the *gehen*-middle must be accounted for, especially given the absence of the silent generic operator (Gen) typically posited for canonical middles (Lekakou 2005; Schäfer 2008). We propose that a generic reading emerges as a byproduct of the lack of Voice and its modal semantics. A thematic Voice head links the Agent to the specific episodic event via Event Identification (Kratzer 1996). However, this is not possible as *gehen* selects a vP. Consequently, the agentive semantics entailed by the encyclopedic knowledge of the root cannot be mapped to a specific event. This unmapped Agent interacts directly with the existential modal operator introduced by *gehen*. The modal operator can only evaluate the dispositional feasibility of the target state, rather than anchoring an event to a syntactically encoded Agent. Therefore, the unmapped implicit Agent receives an arbitrary default interpretation. Thus, the generic flavour is an interpretive consequence of scoping a modal operator over an unaccusative configuration built from an encyclopedically agentive root.

Overall, the *gehen*-middle represents a tightly restricted configuration defined by a constellation of interacting formal characteristics. Given that German already possesses a productive canonical middle, the emergence of this idiosyncratic periphrasis presents a historical puzzle. Notably, the construction bears some similarity to modal control structures under *gehen* (cf. Holl 2010), which we assume to be its source. If correct, we must ask how and why a biclausal control structure was reanalysed as monoclausal raising structure.

# 6 The historical development of the *gehen*-middle

Previous work on the historical development of non-canonical passives in German has mostly focused on modal infinitives, i.e., [*sein*+*zu*-INFINITIVE] (Behaghel 1923; Ebert 1976; Boon 1981; Demske-Neumann 1994). Eroms (1990) offers brief remarks on other construction types that somewhat resemble the regular passive, but to date, no diachronic studies have dealt explicitly with the *gehen*-middle. We attempt to narrow this gap, focusing on the following research questions:

1. What is the historical origin of the *gehen*-middle (i.e., its earliest attestations and potential sources)?

2. How has the construction developed subsequently with regard to its distribution across verb classes, text types, and syntactic contexts (root vs. embedded clauses)?

3. Which factors governed its emergence and further development?

In connection with the first research question, we would like to pursue the hypothesis that the *gehen*-middle evolved from a reanalysis of (arbitrary) control structures linked to the expression of possibility (after *gehen* had developed a relevant meaning):

(53) possibility reading of ‘to go’ → control structures → raising structures (structural simplification)

Before we flesh out this proposal in more detail, we take a closer look at the relevant diachronic facts, presenting the results of a corpus study carried out in the historical corpora of the *Digitales Wörterbuch der Deutschen Sprache* (DWDS).

## 6.1 The rise and loss of the *gehen*-middle – a corpus study

To trace the historical development of the *gehen*-middle, we conducted a corpus study in the historical subcorpora of the DWDS, which cover a time span from 1465 to 1987 and contain approximately 294 million tokens.[13] We extracted all cases (i) where an infinitive (VVINF/VVIZU) precedes a form of ‘to go’ and (ii) where a form of ‘to go’ precedes an infinitive (max. distance 5 words, with the infinitive at the end of the clause).

The search produced 3900 hits, which were narrowed down manually to 95 relevant examples, spanning the years 1715 to 1931. Interestingly, the number of hits far exceeded those in present-day corpora such as DeReKo, despite the smaller size of the historical corpus. The earliest examples found date from the early eighteenth century, given in (54).

13. We used the DWDS corpora, which cover the early and late Early New High German/early New High German period when the *gehen*-middle likely emerged; searches in the reference corpora of Middle and Early New High German and the GermanC corpus yielded no relevant results.

(54) a. *Heisset ein […] Tüchlein, brennend in das Wein-Faß hängen, welches*
bid a […] handkerchief burning into the wine-barrel hang which.NOM
*zu zapffen gehet, oder nicht voll gefüllet ist, […]*
to tap.INF goes, or not fully filled is, […]
'Put a burning handkerchief in the wine barrel, which can be tapped or is not entirely full'
(Corvinus, Gottlieb Siegmund: Nutzbares, galantes und curiöses Frauenzimmer-Lexicon. Leipzig, 1715)

b. *so bald nun Adversarius pariret haben wird und von der Kling die*
as soon now adversary parried have will and from the blade the
*Quart zu stossen gehe drehet man den Leib in Geschwindigkeit*
quarte.NOM to thrust.INF goes turn one the body in pace
*herum […]*
around […]
'As soon as the opponent has parried and the quarte can be thrusted with the blade, the body should be turned with pace…'
(Doyle, Alexander: Neu Alamodische Ritterliche Fecht- und Schirm-Kunst. Nürnberg u. a., 1715)

As can be seen from Figure 8, there is a gap of around 100 years between the above examples and later cases. Throughout the nineteenth century, the construction remains a low frequency phenomenon (< 0.5 occurrences per million tokens).[14] The bulk of all cases comes from the (late) nineteenth century; thereafter, the construction seems to be on its way out.

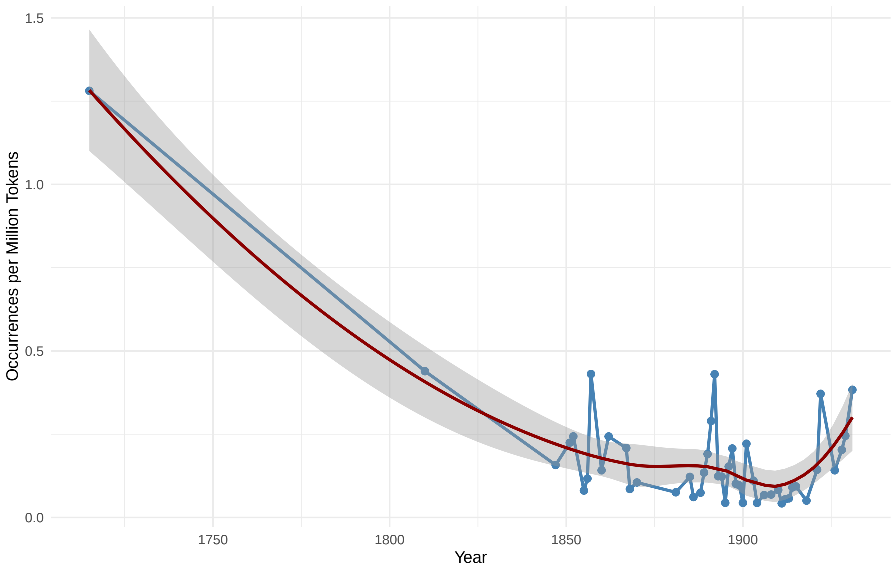



**Figure 8:** Historical Development of the *gehen*-middle (normalized frequencies per year and Loess-smoothed curve)

14. Note that the normalized frequencies for the eighteenth century are misleading – there are only two examples, but the amount of text is much smaller than in the (late) nineteenth century; basically the same goes for the years after 1918.

Taking a closer look at the syntactic contexts in which the *gehen*-middle occurs, it turns out that the construction is not evenly distributed over clause types and verb classes, revealing some similarities but also differences between the historical cases and present-day instantiations of the *gehen*-middle. First, and similar to present-day German (PDG), we can observe an effect of verb class. The majority of all cases involve Accomplishments (69/95 cases, 72.6%). A typical example involving the verb 'to change' is shown in (55).

(55) ***gehen* + Accomplishment**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| *Was* | *einmal* | *geschehen,* | *das* | *geht* | *nicht* | *zu* | *ändern.* |
| what | once | happened | that.NOM | goes | not | to | change.INF |

'What happened once cannot be changed.'
(Deutsches Sprichwörter-Lexikon. vol. 1, 1867, p. 1585)

However, there is also more variation than in PDG, including Achievements as in (56) (20/95 cases, 21.1%) and Activities as in (54a) above (6/95 cases, 6.3%), see Table 2 and Figure 9.

(56) ***gehen* + Achievement**

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| *der* | *[...]* | *Marketender* | *versorgte* | *uns* | *mit* | *Allem,* | *was* | *nur* | *zu* | *kaufen* | *geht* |
| the | [...] | sutler | provided | us | with | everything | what.NOM | only | to | buy.INF | goes |

'The sutler provided us with everything that could be bought.'
(Stolle, Ferdinand (Hrsg.); Diezmann, August (Hrsg.): Die Gartenlaube. Jg. 4, 1856)

| Clause type | Accomplishments | Achievements | Activities | States | Total |
|---|---|---|---|---|---|
| Embedded clauses | 61 (74.4%) | 16 (19.5%) | 5 (6.1%) | 0 | 82 |
| Root clauses | 8 (61.5%) | 4 (30.8%) | 1 (7.7%) | 0 | 13 |
| Total | 69 (72.6%) | 20 (21.1%) | 6 (6.3%) | 0 | 95 |

**Table 2:** Historical instances of the *gehen*-middle – distribution over clause types and verb classes

We also find a strong preference for embedded clauses (83/95 cases, 87.4%), with *gehen* consistently right-adjacent to the *zu*-infinitive, suggesting verb clustering as a precondition for the construction's licensing and grammaticalization. Figure 9 visualizes these distributions, and notably shows that the earliest examples (see 54) involve Activities rather than Accomplishments.

Additional grammatical and extra-grammatical factors that influence the distribution of the construction in the historical data include tense, the presence of adverbs as well as negation, and genre/text type.

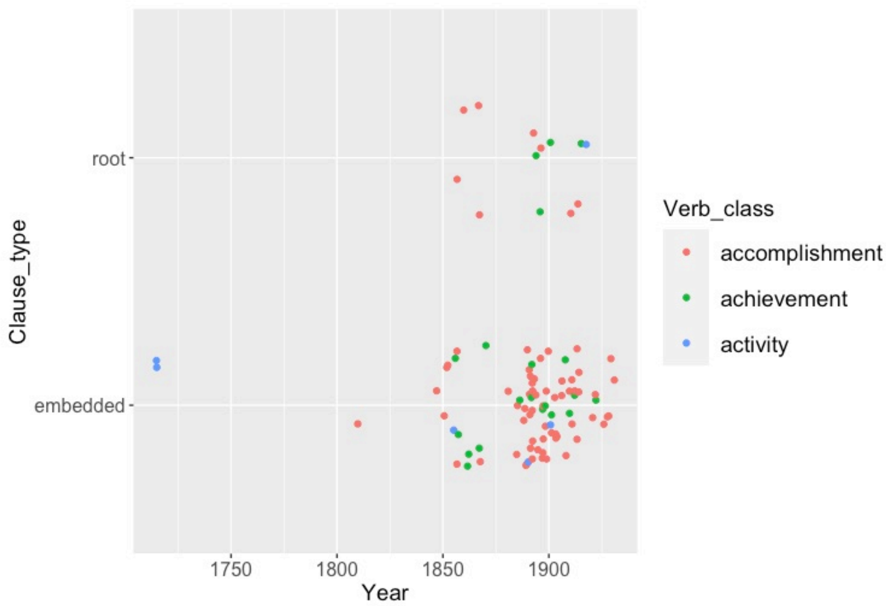



**Figure 9:** The 'go to' passive in the historical DWDS corpora: clause type and verb class

**Tense** The vast majority of examples (73 of 95, 77%) are in the present tense, which is probably a reflex of the generic flavour of the construction. But again, there is more variation in the historical data. In contrast to PDG, the *gehen*-middle is not confined to simple tenses but also shows up with periphrastic verb forms as in (57).

(57) **periphrastic verb form**

| | | | | | |
|---|---|---|---|---|---|
| *er* | *sei* | *aber* | *nicht* | *auszuheben* | *gegangen* |
| he.NOM | be.BE.PRES.SBJ | however | not | out-to-lift.INF | gone |

'However, it [the shutter] couldn't be taken off its hinges'
(Die Ermordung der Witwe Dellbrück, geb. Hahnemann. In: Der neue Pitaval, Bd. 28. Leipzig, 1860)

**Presence of adverbials/negation** Modification by some kind of adverbial expression is the most common pattern (49/95 cases, 51.6%). Some relevant cases are shown in (58). The negation *nicht* 'not' is present in 19/95 cases (20%), see e.g., (55) above for illustration.

(58) a.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| *Da* | *diese* | *letzteren* | *am* | *leichtesten* | *abzuändern* | *gehen* | *[...]* |
| since | the | latter.PL | on-the | easiest | modify.INF | go.PL | |

'Since the latter are the easiest to modify...'
(Theodor Pregél, Die Ermittelung der Stufenscheibenverhältnisse für Werkzeugmaschinen. In: Dinglers Polytechnisches Journal, 1893/287, pp. 248-252. Stuttgart, 1893)

b. *Daß es auch billiger zu machen geht, als die Regierung vorschlägt,*
that it.NOM also cheaper to make.INF goes than the government proposes
*das beweist uns die Schweiz [...].*
that proves us the Switzerland
'That it can also be done more cheaply than the government proposes is demonstrated by Switzerland.'
(Verhandlungen des Reichstages. Berlin, 1899.)

However, a surprisingly large portion of the historical examples lack a modifying element as in (59) (27 cases, 28.4%), which is not expected given the PDG facts, where the presence of adverbials or negation seems to be compulsory.

(59) *die [...] einen Schwiegersohn wünschte, der zu lenken und*
who.FEM [...] a son-in-law wished who.NOM.MASC to steer.INF and
*leiten [...] ging*
lead.INF [...] went
'who wished for a son-in-law who could be steered and led'
(Keil, Ernst (Hrsg.): Die Gartenlaube. Jg. 3 (1855))

**Impact of genre/text type I** The vast majority of all occurrences appear in utility prose and science (subgroups according to the DWDS corpus; manually added: 'science' for papers in *Dinglers Polytechnisches Journal*, one of the foremost technical journals in nineteenth century Germany). The distribution over different genres is visualized in Figure 10 (absolute numbers of occurrences and normalized frequencies per million words).

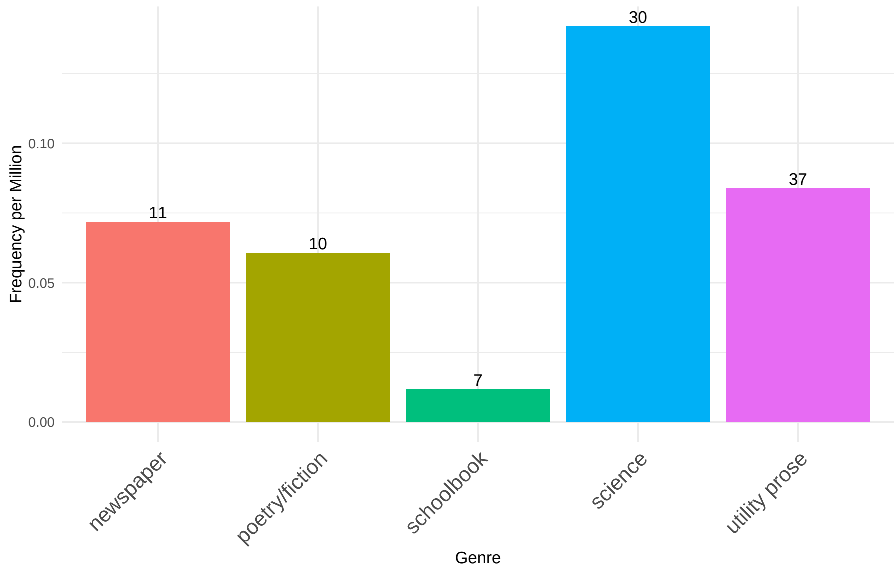



**Figure 10:** Distribution of the *gehen*-middle across text types (normalized frequencies and absolute numbers)

**Impact of genre/text type II** There is a noteworthy spike in texts linked to the genre 'science' in the late nineteenth century (while examples in other types of utility prose are more evenly distributed), compare Figure 11. All relevant examples (27 cases) appeared in *Dinglers Polytechnisches Journal*. A typical example of this type of technical writing is given in (60).

(60) *so dass die mit äusserem Schraubengewinde versehene Bohrspindel leicht*
so that the with external screw-thread provided drill-spindle.NOM easy
*einzubringen und herauszunehmen geht*
to-insert.INF and to-remove.INF goes
'so that the drill spindle, which is provided with an external screw thread, can be easily inserted and removed'
(Neuerungen in der Tiefbohrtechnik von Eugen Gad in Darmstadt. In: Dinglers Polytechnisches Journal (Hg. A. Hollenberg), Jg. 1890/278, S. ad-156. Stuttgart, 1890)

However, upon closer inspection, the sudden increase could in fact reflect individual stylistic preferences, since at least 10 of all 27 cases appear in articles by Theodor Pregél, a professor of engineering at the University of Chemnitz, see (61) for another example reflecting this kind of style. Regardless, the use of the *gehen*-middle in text types such as science/technical writing is not expected, given its present-day link with colloquial speech.

(61) *An einer Ringleiste des Rundtisches ist der Schneckenkranz h angebracht,*
at a ring-rail of-the rotary-table is the worm-gear h mounted
*welcher durch die Schnecke i zu betreiben geht.*
which.NOM through the worm i to drive.INF goes
'At a ring rail of the rotary table, the worm gear h is mounted, which can be driven by the worm i.'
(Th. Pregél: J. E. Reinecker's Werkzeugmaschinen. In: Polytechnisches Journal (Hg. W. Pickersgill), Jg. 1901/316, S. 357-361. Stuttgart, 1901.)

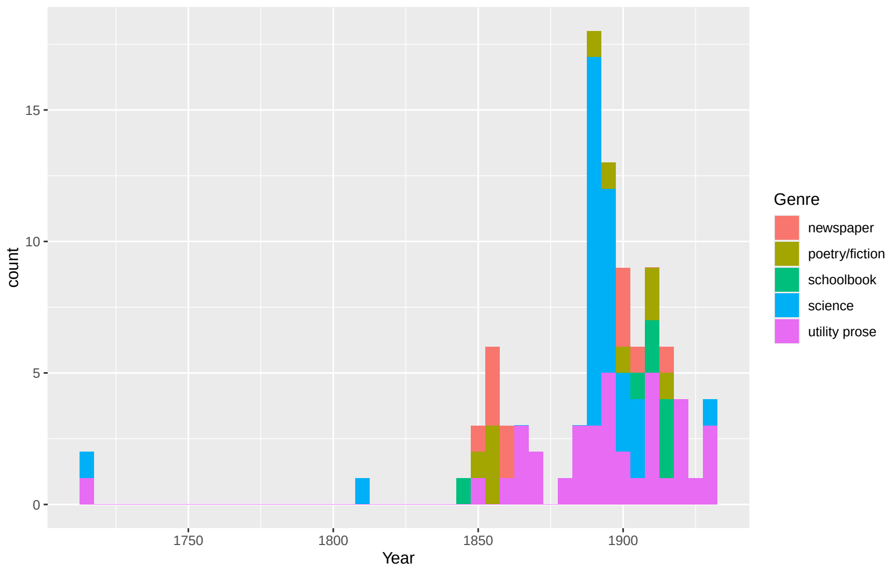



**Figure 11:** Proportion of text types per 10 years

The major empirical results of the corpus study are as follows: From the data we have gathered, the *gehen*-middle appears to have evolved rapidly between the eighteenth and nineteenth centuries (recall that the construction does not seem to exist in earlier historical stages of German). We have also noted a strong tendency for *gehen* to be right-adjacent to the infinitive, which suggests that verb cluster formation was a necessary precondition for the grammaticalization of the *gehen*-middle. The construction reached its prime in the late nineteenth century (in certain text types, technical writing, in particular); at that time, it had a wider distribution across verb classes, verb forms, and non-colloquial registers. Afterwards, the *gehen*-middle seems to be in decline, at least in the written language. Interestingly, the increasing restriction to colloquial registers appears to go hand in hand with growing grammatical constraints, a development that ultimately excluded periphrastic verb forms and Achievements from the modern construction.

## 6.2 The origin of the *gehen*-middle

In many languages, passive auxiliaries can be traced back to lexical verbs. Relevant processes of grammaticalization frequently affect verbs expressing inchoative or change-of-state meanings (Haspelmath 1990; Heine and Kuteva 2002; Wiemer 2011). Well-known examples include 'get/receive' (English, ancient Chinese), 'become' (German), and 'give' (Luxembourgish and Moselle Franconian dialects, Nübling 2006; Lenz 2007). Another likely source are motion verbs, in particular, 'come' (in the so-called 'Alpine passive', Wiesinger 1989; Ramat 1998; Gaeta 2018) and 'go' (Urdu, Quechua, Scottish Gaelic, Alpine and colloquial varieties of German).[15]

15. Note that in historical and colloquial/dialectal varieties of German, auxiliary uses of 'go' and 'become' seem to fulfill a similar range of functions, including passive formation and expression of future (see e.g., Demske 2020:on 'go' in ENHG).

In what follows, we take a closer look at the diachronic scenario that led to the emergence of the *gehen*-middle. More precisely, we propose that the relevant grammaticalization process goes back to a conspiracy of semantic and syntactic changes, involving (at least) the following set of ingredients:

1. A semantic change in which 'to go' turns into an element expressing possibility;

2. Analogical extension of 'go' expressing possibility to contexts in which it combines with a clausal infinitive (a control structure, with purpose clauses as a potential model/facilitating factor);

3. Reanalysis of the control infinitive as a raising structure (necessarily linked to the availability of ambiguous bridging contexts and another semantic change in which semi-modal 'go' loses its capability to assign a thematic role to its subject; (see Traugott 1993; Heine and Miyashita 2008 on 'threaten' and 'promise' in the history of English and German).

### 6.2.1 Semantic change: development of a (im)possibility reading

As part of the core vocabulary, the verb 'go' is typically of considerable antiquity and has developed a wide range of additional, often figurative meanings across languages. This is also true of German *gehen*, which is a massively polyfunctional/polysemic lexical element (cf. the huge entry in Grimm's German Dictionary (DWB)). In one of the potential figurative readings, 'go' expresses possibility or feasibility as e.g., in *Das geht nicht* 'That is impossible (lit. that goes not)'. Possibility-'go' may take a nominal or clausal (infinitival) argument. This is shown in (62) and (63), respectively.

(62) *In ein Gefäss von fünf Litern Inhalt gehen keine 6 1/2 Liter Wein, schon*
in a container of five liters content go no 6 1/2 liter wine even
*gar nicht, wenn dazu noch 3 kg Zwetschgen und 1,8 kg*
at-all not if there-to also 3 kilograms plums and 1.8 kilograms
*Zucker darin Platz haben sollten. Das geht nicht, wird nie gehen, solange*
sugar there-in place have should that goes not will never go as-long-as
*Zwetschgen auf Bäumen wachsen.*
plums on trees grow

'A five-litre container cannot hold six and a half litres of wine—let alone if it is also supposed to accommodate three kilograms of plums and 1.8 kilograms of sugar. That is impossible and will never be possible, as long as plums grow on trees.'
(A97/SEP.24330 St. Galler Tagblatt, 16.09.1997, Ressort: TB-LBN (Abk.); Zuviel des Guten)

(63) *Sich zurückzulehnen und darauf zu warten, was die Politik tut, geht nicht mehr.*
REFL to-sit-back and there-on to wait what the politics do goes not no-longer
'It is no longer possible to sit back and wait to see what politicians will do.'
(NUZ17/DEZ.01471 Nürnberger Zeitung, 19.12.2017, S. 12)

According to the DWB, 'go' came to be used to express possibility/feasibility from the 16th century on. Two early examples with a nominal and a clausal complement are given in (64) and (65), respectively.

(64) *Nein/ nein. Das gehet nicht.*
no no that goes not
'No, no. That is impossible.'
(DWDS; Fleming, Paul: Teütsche Poemata. Lübeck, 1642)

(65) *Doch daß man dem gerichte Die schuld aufflegen wil; geht nicht.*
however that one the.DAT court.DAT the blame on-lay want goes not
'However, it is not possible to lay the blame on the court.'
(DWDS; Gryphius, Andreas: Teutsche Reim-Gedichte. Frankfurt (Main), 1650)

The DWB mentions two potential sources of the possibility reading for 'go'. First, a related interpretation is linked to the particle verb *angehen* 'on-go', which originally expressed an inchoative meaning ('to start'), also in connection with plant growth ('to take roots, sprout'). From this, a possibility reading was derived, either with a nominal subject as in (66), or an infinitival subject clause (typically an arbitrary control structure) as in (67):

(66) *Der Käyser schüttelte den Kopff/ und sagte: Nein/ ihr redliche und Edle Nordische Helden/ das gehet nicht an!*
the emperor shook the head and said no you honest and noble Nordic heroes that goes not on
'The emperor shook his head and said: No, you honest and noble Nordic heroes, that won't do!'
(DWDS; Schupp, Johann Balthasar: Schrifften. Hrsg. v. Anton Meno Schupp. [Hanau], 1663)

(67) *den Bibliothekfonds einfach mehr zu belasten ging allerdings nicht an,* [...]
the.ACC library-fund simply more to strain went however not on, [...]
'However, simply putting more strain on the library fund was not an option.'
([N. N.]: Verhandlungen des Reichstages. Berlin, 1895)

Another potential source is a figurative meaning of 'go' in connection with directional adverbials, expressing that two things/workpieces potentially fit together as in (68).

(68) *das man einen eisern ring daran schieben kan, der geheb daran gehet*
that one an iron ring there-on slide can that.NOM.MASC exactly there-on goes
'that you can slide an iron ring on it that fits exactly'
(DWB citing Erker 8a, Frankfurt/M., 1580)

### 6.2.2 Analogical extension: combination with control infinitives

Cases such as (67) above where an instance of *angehen* 'on-go' that expresses a possibility/feasibility reading is combined with a control infinitive are robustly attested in the historical subcorpora of the DWDS.[16] What we would like to propose is that this pattern provided a model for the analogical extension of the use contexts for simple possibility-'go', which as a result came to select a control infinitive as in (69) (realized as its subject; cf. Haspelmath 1989 on the grammaticalization of the *zu*-infinitive and its compatibility with matrix predicates expressing possibility).

(69) *...um zu hören, ob es ginge, mit ihr zusammen zu reisen*
in-order to hear if it went.PRET.SBJ with her together to travel
'...to hear if it would be possible to travel with her'
(Varnhagen von Ense, Rahel: Rahel. Ein Buch des Andenkens für ihre Freunde. Bd. 2. Berlin, 1834)

Note that in (69), the clausal infinitive has undergone extraposition. In PDG, this is usually taken to be a diagnostic for a so-called incoherent construction, in which no verb cluster formation has taken place and in which the infinitival complement represents a control structure (Bech 1955/1983; Grewendorf 1988; Haider 2010).

### 6.2.3 Reanalysis of control infinitives as raising structures

The next and final step that led to the emergence of a new passive-like construction involved a reanalysis of ambiguous control infinitives as raising structures, accompanied (or facilitated) by further semantic bleaching of (then semi-modal) 'go', which lost its capability to assign a thematic role to its subject (similar to other verbs in the recorded history of the Germanic languages such as 'seem', 'threaten', or 'promise', cf. Traugott 1993; Heine and Miyashita 2008).[17]

This transition was possibly fuelled by independent changes that led to an auxiliarization of 'go' in ENHG (cf. e.g., Demske 2020).[18] Moreover, recent work on raising and control infinitives in ENHG by Demske (2015), De Cesare (2021), and De Cesare and Demske (2025) shows that the different syntactic behaviour of raising and control infinitive (e.g., concerning

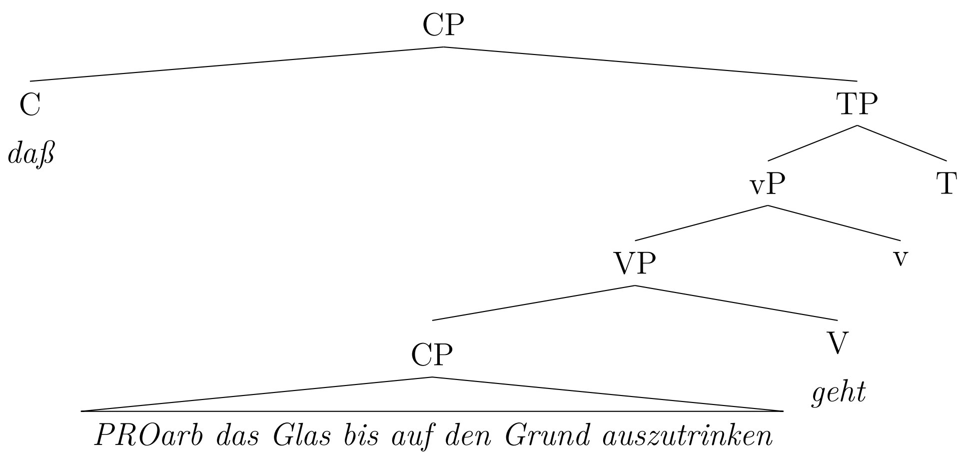
16. A sample search query "geht nicht an #5 zu $p=VVINF" yielded 64 relevant hits between 1778 and 1932.
17. More generally, it seems that control infinitives are a likely source of raising constructions across languages.
18. Another potentially promoting factor is the existence of other passive-like infinitival constructions such as the modal *sein+zu*-passive, which developed in the ENHG period, Demske-Neumann cf. 1994:

extraposition of the infinitival complement) is a recent development from the eighteenth century on, lending further support to the idea that control infinitives could easily be misparsed as raising structures. Potential bridging contexts are intraposed arbitrary control infinitives in which case marking of the internal argument was not distinctive (due to common NOM/ACC syncretism in German) as in (70).

(70) *daß das Glas bis auf den Grund auszutrinken geht*
that the glass.NOM to on the bottom to-drink.INF goes
'that the glass can be drunk to the bottom'
(Pechner, Fr.: Handbuch für Lehrer beim Gebrauche des Preussischen Kinderfreundes. Der gesammte deutsche Sprachunterricht in Volksschulen oder die Uebungen im Lesen, der Grammatik, Orthographie und dem mündlichen und schriftlichen Gedankenausdrucke [...]. Königsberg, 1847)

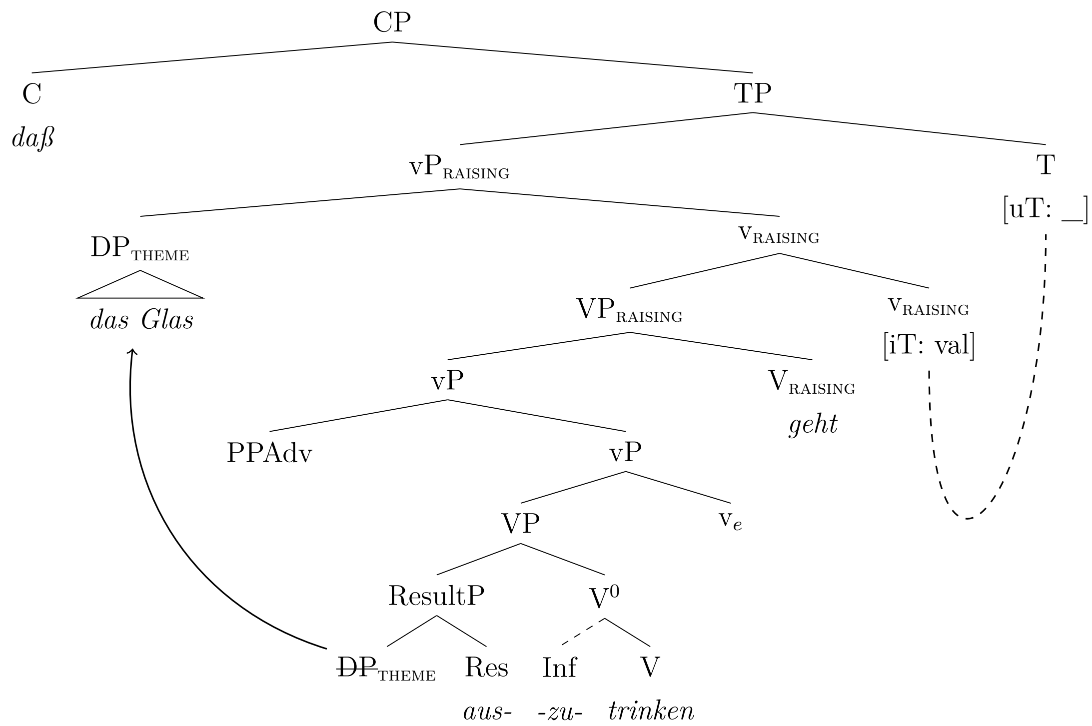
The reanalysis of a control structure as raising infinitive leads to structural simplification in that an originally biclausal structure (involving two CPs) is reduced to a coherent verb cluster consisting of two extended verbal projections. This change is illustrated in the difference between (71) and the reanalysed structure in (72).

(71)




(i) *Uebers Jahr kann man sehen, ob's wird [zu schelten sein] oder [zu loben gehen].*
over-the year can one see whether=it.NOM will to reproach be or to praise.INF go.INF
'Over the course of the year, one can see whether it will turn out to deserve reproach or praise.'
(Wander, Karl Friedrich Wilhelm (Hrsg.): Deutsches Sprichwörter-Lexikon. Bd. 2. Leipzig, 1870)

(72)




The loss of a fully biclausal structure is accompanied by a number of additional structural changes. First, 'go' turns into a raising verb that does not assign a thematic role to its subject and selects a verbal complement with minimal functional structure (and thus cannot license PRO). The infinitival complement (formerly a CP that functions as the subject of the matrix clause) is reduced to a (unaccusative) vP that loses its capability to assign accusative case; consequently, its internal argument undergoes raising to the higher TP where it receives nominative case.

As a result of this reanalysis, genuine examples of the *gehen*-middle should lack extraposition of the infinitival complement, since raising (and other clause union) constructions require verb clustering in German (cf. Haider 2010). This is in line with our observation that the *zu*-infinitive always directly precedes finite 'go' in all cases, where the *gehen*-middle is part of an embedded clause.[19] Moreover, we expect new raising patterns to appear in which *gehen* clearly combines with a DP subject. Relevant diagnostics include (i) unambiguous nominative marking on the subject DP and (ii) plural agreement on *gehen*. Some relevant examples are given in (73) and (74).

(73) *…diesen entschiedenen Gegensatz, welcher sichtbar und empfindbar ist und*
…this clear opposition which visible and perceptible is and

***der*** *nicht aufzuheben geht*
that.MASC.NOM not eliminated.INF goes

'this clear opposition, which is visible and perceptible, and which cannot be eliminated'

19. But note that the loss of extraposition in raising structures was probably an independent, slightly earlier or parallel development (cf. Demske 2015; De Cesare 2021; De Cesare and Demske 2025).

(Johann Wolfgang Goethe, Zur Farbenlehre, 1810)

(74) *Da diese letzteren am leichtesten abzuändern* ***gehen*** […]
since the latter.PL on-the easiest modify.INF go.PL
'Since the latter are the easiest to modify...'
(Theodor Pregél, Die Ermittelung der Stufenscheibenverhältnisse für Werkzeugmaschinen. In: Dinglers Polytechnisches Journal, 1893/287, pp. 248-252. Stuttgart, 1893)

Hence, we should expect an increase in cases that clearly involve raising (relative to the number of cases that are ambiguous between raising and control). This expectation is, however, only borne out for the short period in the late nineteenth century, where the use of the *gehen*-middle was apparently more common (see Figure 12). The absence of a clear-cut pattern may also be attributable to the limited size of the data set (n=95). Regardless, additional data from the eighteenth to the early twentieth century is necessary for a more robust understanding of the phenomenon.

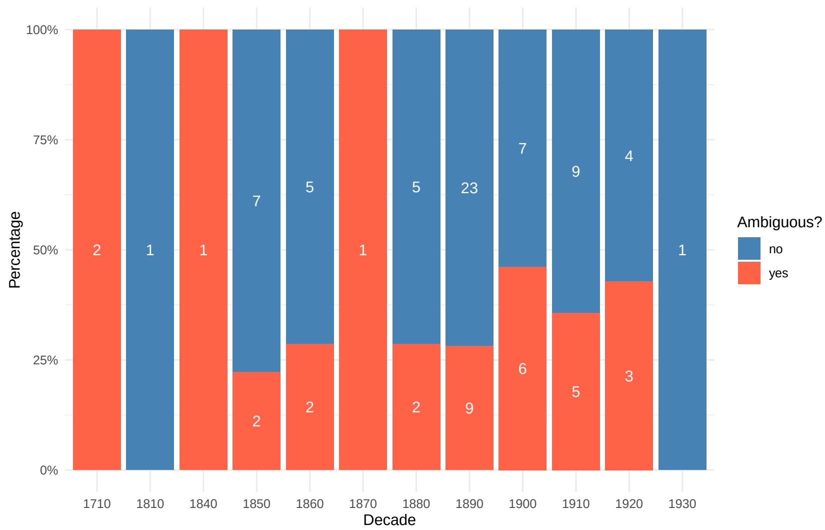



**Figure 12:** Proportion of ambiguous occurrences over time (per decade with absolute numbers)

# 7 Concluding summary

The first part of this paper presented a novel in-depth investigation of [*gehen*+*zu*-INFINITIVE]. Using experimental methodology and limited qualitative *post-hoc* syntactic and semantic testing, we were able to confidently classify [*gehen*+*zu*-INFINITIVE] as an infinitival construction expressing middle voice. It is neither an infinitival passive (*contra* Holl 2010; Abraham 2020; Müller 2019) nor a periphrastic (Impossibility) anticausative (*contra* Cysouw 2023). The behaviour of this middle is determined by constraints involving argument structure, aspect, and modal scope. First, *gehen* is a semi-modal (possibility) unaccusative *v*-raising head, but lacks any eventive or thematic properties itself. Second, this head only selects an eventive vP, ruling out Agent introduction in a lower VoiceP. Aspectual restrictions are instead enforced at

C/I and LF. As a modal operator, *gehen* requires a structurally distinct target state to evaluate. We argued that only causative vP-configurations can provide this environment, via merge of a complement ResultP. Together, these factors account for the passive-like interpretation, the dispositional properties of the promoted argument, and the lexical-semantic restriction to Accomplishments. Conversely, the semantic and syntactic properties associated with Achievements and Activities fail to project an aspectually compatible result state, causing the derivation to fail at the interfaces.

The second part of the paper dealt with the historical development of the *gehen*-middle. Based on a study in the historical subcorpora of the DWDS corpus it was shown that the construction presumably developed in the early eighteenth century via a reanalysis of ambiguous control structures as raising infinitives, which gave rise to a (semi-modal) raising version of 'go'. This structural shift was made possible by an earlier semantic change in which 'go' came to express possibility/feasibility. However, we have also seen that the heyday of the new pattern was rather short-lived, giving rise to the impression of a failed change. Possibly, this has to do with the fact that the *gehen*-middle has never lost its colloquial character (which initially might have contributed to its attractiveness), which ultimately led to its demise in the written language.

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# A Appendix: Raw Data

## Data for: Compound Tense

**Table 3:** Descriptive Statistics: Compound Tense

| **Condition** | **Item (Sentence)** | $M$ | $SD$ | $M_z$ | $SD_z$ |
|---|---|---|---|---|---|
| scheinen_past | Die Journalistin schien gut informiert zu sein. | 4.92 | 0.33 | 1.39 | 0.41 |
| gehen_past | Der Wagen ging nicht zu reparieren. | 3.51 | 1.39 | 0.42 | 0.69 |
| gehen_past | Der Wagen ist nicht zu reparieren gegangen. | 1.44 | 0.77 | -0.95 | 0.52 |
| scheinen_past | Die Journalistin hat gut informiert zu sein geschienen. | 1.47 | 0.87 | -0.94 | 0.62 |

## Data for: Extraposition

**Table 4:** Descriptive Statistics: Extraposition

| **Condition** | **Item (Sentence)** | $M$ | $SD$ | $M_z$ | $SD_z$ |
|---|---|---|---|---|---|
| gehen_!extraposition | Louisa ist sehr glücklich, dass das Spiel endlich zu installieren geht . | 3.31 | 1.15 | 0.30 | 0.56 |
| gehen_extraposition | Louisa ist sehr glücklich, dass das Spiel endlich geht zu installieren. | 1.31 | 0.62 | -1.03 | 0.44 |

## Data for: Datives and Double Arguments

**Table 5:** Descriptive Statistics: Datives and Double Arguments

| **Condition** | **Item (Sentence)** | $M$ | $SD$ | $M_z$ | $SD_z$ |
|---|---|---|---|---|---|
| sein_dat | Ihm ist nicht zu helfen. | 4.81 | 0.64 | 1.32 | 0.56 |
| gehen+distransitive | Die Email geht ihm nicht zu schicken. | 1.17 | 0.50 | -1.14 | 0.37 |
| gehen_dat | Ihm geht nicht zu helfen. | 1.38 | 0.68 | -0.99 | 0.45 |

# Data for: Agentive von-phrases and Implicit für-phrases

**Table 6:** Descriptive Statistics: Agentive von-phrases and Implicit für-phrases

| **Condition** | **Item (Sentence)** | $M$ | $SD$ | $M_z$ | $SD_z$ |
|---|---|---|---|---|---|
| werden+von | Das Schmetterlingshaus wird von allen Besuchern geliebt. | 4.78 | 0.51 | 1.30 | 0.46 |
| sind+für | Die Fresken sind jetzt wieder für alle Besucher zu sehen. | 4.69 | 0.52 | 1.23 | 0.40 |
| lassen+von | Der Bürgermeister lässt eine Mauer von den Bauarbeitern errichten. | 4.32 | 0.95 | 0.97 | 0.54 |
| gehören+von | Dieser laute Krach gehört von der Polizei verboten. | 3.89 | 1.11 | 0.68 | 0.67 |
| middle+für | Das Buch liest sich für sie gut. | 3.65 | 1.24 | 0.54 | 0.77 |
| gehen+für | Der Tisch ging für den Lehrer einfach abzuwischen, aber nicht für die Kinder. | 2.92 | 1.36 | 0.03 | 0.73 |
| sind+von | Die Fresken sind jetzt wieder von allen Besuchern zu sehen. | 2.88 | 1.27 | 0.03 | 0.82 |
| gehen+für | Die Tür geht für Anna kaum aufzuschließen. | 2.76 | 1.32 | -0.06 | 0.68 |
| gehen+für | Der Blinker geht für Younes leicht zu reparieren. | 2.58 | 1.22 | -0.19 | 0.59 |
| gehen+von | Der Tisch ging vom Lehrer einfach abzuwischen, aber nicht von den Kindern. | 2.12 | 1.19 | -0.49 | 0.63 |
| gehen+von | Die Tür geht von Anna kaum aufzuschließen. | 1.81 | 1.03 | -0.69 | 0.57 |
| gehen+von | Der Blinker geht von Younes leicht zu reparieren. | 1.62 | 0.83 | -0.83 | 0.43 |
| middle+von | Das Buch liest sich von ihr gut. | 1.47 | 0.87 | -0.92 | 0.58 |

# Data for: Instrumental adjuncts

**Table 7:** Descriptive Statistics: Instrumental adjuncts

| **Condition** | **Item (Sentence)** | $M$ | $SD$ | $M_z$ | $SD_z$ |
|---|---|---|---|---|---|
| gehen+instrument | Die Schraube geht aber mit einem besonderen Schraubenzieher leicht zu lösen. | 3.21 | 1.35 | 0.21 | 0.75 |

# Data for: Adverbial Modification

**Table 8:** Descriptive Statistics: Adverbial Modification

| **Condition** | **Item (Sentence)** | $M$ | $SD$ | $M_z$ | $SD_z$ |
|---|---|---|---|---|---|
| gehen+otheradv | Die Tür geht kaum aufzuschließen. | 3.85 | 1.27 | 0.64 | 0.70 |
| gehen+midadv | Die Tür geht jetzt leicht aufzuschließen. | 3.39 | 1.20 | 0.35 | 0.58 |
| gehen+midadv | Der Tisch ging dann ganz einfach abzuwischen. | 3.46 | 1.36 | 0.39 | 0.70 |
| gehen+otheradv | Die Tür geht jetzt wieder aufzuschließen. | 3.43 | 1.45 | 0.37 | 0.74 |
| gehen+nicht | Die Schraube geht nicht einfach zu lösen. | 3.25 | 1.33 | 0.25 | 0.69 |
| gehen+nicht | Der gefrorene Apfel ging nicht zu essen. | 3.08 | 1.32 | 0.14 | 0.48 |
| gehen+otheradv | Die Tür geht jetzt aufzuschließen. | 2.99 | 1.38 | 0.07 | 0.72 |
| gehen+otheradv | Der Tisch ging zuerst nicht abzuwischen. | 2.71 | 1.31 | -0.10 | 0.66 |
| gehen+otheradv | Der Quark geht noch zu essen. | 2.25 | 1.20 | -0.40 | 0.66 |

## Data for: Vendler's Verb Classes

**Table 9:** Descriptive Statistics: Vendler's Verb Classes

| Class | Condition | Item (Sentence) | *M* | *SD* | $M_z$ | $SD_z$ |
|---|---|---|---|---|---|---|
| Accomplishment | gehen+midadv | DieTür geht jetzt leicht aufzuschließen. | 3.76 | 1.24 | 0.58 | 0.63 |
| Accomplishment | gehen_past | Der Wagen ging nicht zu reparieren. | 3.51 | 1.39 | 0.42 | 0.69 |
| Accomplishment | gehen+midadv | Der Tisch ging dann ganz einfach abzuwischen. | 3.46 | 1.37 | 0.39 | 0.70 |
| Accomplishment | gehen+nicht | Die Schraube geht nicht einfach zu lösen. | 3.25 | 1.34 | 0.25 | 0.69 |
| Accomplishment | gehen+nicht | Das alte Spiel geht nicht zu installieren. | 3.21 | 1.30 | 0.22 | 0.67 |
| Accomplishment | gehen+nicht | Das geht leider nicht zu ändern. | 2.94 | 1.35 | 0.06 | 0.76 |
| Achievement | gehen+nicht | Der Schaffner geht nicht zu überzeugen. | 1.75 | 0.90 | -0.72 | 0.49 |
| Achievement | gehen+nicht | Der Berggipfel geht einfach nicht zu erreichen. | 1.78 | 0.86 | -0.71 | 0.46 |
| Achievement | gehen+nicht | Robert geht einfach nicht zu schlagen. | 1.81 | 0.99 | -0.68 | 0.51 |
| Achievement | gehen+nicht | Superman geht nicht zutöten. | 2.12 | 1.20 | -0.49 | 0.63 |
| Achievement | gehen+nicht | Die Schwierigkeiten gingen nicht zu überwinden. | 2.17 | 1.06 | -0.45 | 0.55 |
| Activity | gehen+adv | Jimmys Gitarre geht leicht zu spielen. | 3.39 | 1.34 | 0.34 | 0.74 |
| Activity | gehen+otheradv | Der Quark geht noch zu essen. | 2.25 | 1.21 | -0.40 | 0.66 |
| Activity | gehen+nicht | Der gefrorene Apfel ging nicht zu essen. | 2.61 | 1.19 | -0.17 | 0.61 |

# Appendix: Experimental Stimuli and Descriptive Statistics

**Table 10:** Descriptive Statistics for All Experimental Items

| Condition | Item (Original & Translation) | *M* | *SD* | $M_z$ | $SD_z$ |
|---|---|---|---|---|---|
| 1sg | Ich gehe nicht zu ändern? Warum lässt Carla mich nicht einfach so wie ich bin?! *'I can't be changed? Why doesn't Carla just leave me the way I am?!'* | 1.56 | 0.75 | -0.87 | 0.41 |
| 2sg | Du gehst einfach nicht zu ändern. *'You simply cannot be changed.'* | 1.26 | 0.67 | -1.07 | 0.46 |
| Probe_gut | Der Käsekuchen wurde von meinem Bruder aufgegessen. *'The cheesecake was eaten up by my brother.'* | 4.68 | 0.62 | 1.21 | 0.46 |

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Table 10 – continued from previous page

| **Condition** | **Item (Original & Translation)** | $M$ | $SD$ | $M_z$ | $SD_z$ |
|---|---|---|---|---|---|
| Probe_schlecht | Von meinem Brüdern wurde aufgegesst den Käsekuchen. *'By my brothers was eaten up the cheese-cake. (ungrammatical)'* | 1.03 | 0.17 | -1.25 | 0.44 |
| gehen_!extraposition | Louisa ist sehr glücklich, dass das Spiel endlich zu installieren geht . *'Louisa is very happy that the game can finally be installed. (intraposed)'* | 3.29 | 1.40 | 0.30 | 0.74 |
| gehen_!zu | Der Wagen ging nicht reparieren. *'The car could not repair. (missing zu)'* | 1.15 | 0.49 | -1.13 | 0.44 |
| gehen_dat | Ihm geht nicht zu helfen. *'He cannot be helped. (gehen)'* | 1.38 | 0.68 | -0.99 | 0.41 |
| gehen_extraposition | Louisa ist sehr glücklich, dass das Spiel endlich geht zu installieren. *'Louisa is very happy that the game can finally be installed. (extraposed)'* | 1.29 | 0.76 | -1.03 | 0.52 |
| gehen_past | Der Wagen ging nicht zu reparieren. *'The car could not be repaired.'* | 3.51 | 1.39 | 0.42 | 0.72 |
| gehen_past | Der Wagen ist nicht zu reparieren gegangen. *'The car has not been able to be repaired. (compound)'* | 1.43 | 0.73 | -0.95 | 0.45 |
| gehen+adv | Jimmys Gitarre geht leicht zu spielen. *'Jimmy's guitar can easily be played.'* | 3.39 | 1.18 | 0.34 | 0.56 |
| gehen+distransitive | Die Email geht ihm nicht zu schicken. *'The email cannot be sent to him.'* | 1.17 | 0.50 | -1.14 | 0.41 |
| gehen+für | Die Tür geht für Anna kaum aufzuschließen. *'The door can barely be unlocked for Anna.'* | 2.76 | 1.32 | -0.06 | 0.68 |
| gehen+für | Der Blinker geht für Younes leicht zu reparieren. *'The indicator can easily be repaired for Younes.'* | 2.75 | 1.32 | -0.09 | 0.73 |

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Table 10 – continued from previous page

| Condition | Item (Original & Translation) | *M* | *SD* | $M_z$ | $SD_z$ |
|---|---|---|---|---|---|
| gehen+für | Der Tisch ging für den Lehrer einfach abzuwischen, aber nicht für die Kinder. *'The table could simply be wiped off for the teacher, but not for the children.'* | 2.74 | 1.25 | -0.10 | 0.61 |
| gehen+instrument | Die Schraube geht aber mit einem besonderen Schraubenzieher leicht zu lösen. *'The screw can however easily be loosened with a special screwdriver.'* | 3.21 | 1.35 | 0.21 | 0.75 |
| gehen+midadv | Der Tisch ging dann ganz einfach abzuwischen. *'The table could then be wiped off quite simply.'* | 3.46 | 1.36 | 0.39 | 0.70 |
| gehen+midadv | Die Tür geht jetzt leicht aufzuschließen. *'The door can easily be unlocked now.'* | 3.39 | 1.20 | 0.35 | 0.58 |
| gehen+nicht | Das alte Spiel geht nicht zu installieren. *'The old game cannot be installed.'* | 3.65 | 1.46 | 0.49 | 0.75 |
| gehen+nicht | Das geht leider nicht zu ändern. *'Unfortunately, that cannot be changed.'* | 3.57 | 1.36 | 0.44 | 0.69 |
| gehen+nicht | Die Schraube geht nicht einfach zu lösen. *'The screw cannot simply be loosened.'* | 3.25 | 1.33 | 0.25 | 0.69 |
| gehen+nicht | Der gefrorene Apfel ging nicht zu essen. *'The frozen apple could not be eaten.'* | 3.08 | 1.32 | 0.14 | 0.48 |
| gehen+nicht | Der Schaffner geht nicht zu überzeugen. *'The conductor cannot be convinced.'* | 1.94 | 1.09 | -0.58 | 0.66 |
| gehen+nicht | Die Schwierigkeiten gingen nicht zu überwinden. *'The difficulties could not be overcome.'* | 1.83 | 0.95 | -0.67 | 0.56 |
| gehen+nicht | Der Berggipfel geht einfach nicht zu erreichen. *'The mountain peak simply cannot be reached.'* | 1.79 | 0.99 | -0.69 | 0.55 |
| gehen+nicht | Robert geht einfach nicht zu schlagen. *'Robert simply cannot be beaten.'* | 1.62 | 0.88 | -0.79 | 0.53 |
| gehen+nicht | Superman geht nicht zu töten. *'Superman cannot be killed.'* | 1.47 | 0.77 | -0.89 | 0.50 |

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Table 10 – continued from previous page

| **Condition** | **Item (Original & Translation)** | $M$ | $SD$ | $M_z$ | $SD_z$ |
|---|---|---|---|---|---|
| gehen+otheradv | Die Tür geht kaum aufzuschließen.<br>*'The door can barely be unlocked.'* | 3.85 | 1.27 | 0.64 | 0.70 |
| gehen+otheradv | Die Tür geht jetzt wieder aufzuschließen.<br>*'The door can be unlocked again now.'* | 3.43 | 1.45 | 0.37 | 0.74 |
| gehen+otheradv | Die Tür geht jetzt aufzuschließen.<br>*'The door can be unlocked now.'* | 2.99 | 1.38 | 0.07 | 0.72 |
| gehen+otheradv | Der Tisch ging zuerst nicht abzuwischen.<br>*'The table could not be wiped off at first.'* | 2.71 | 1.31 | -0.10 | 0.66 |
| gehen+otheradv | Der Quark geht noch zu essen.<br>*'The quark can still be eaten.'* | 2.25 | 1.20 | -0.40 | 0.66 |
| gehen+von | Der Tisch ging vom Lehrer einfach abzuwischen, aber nicht von den Kindern.<br>*'The table could simply be wiped off by the teacher, but not by the children.'* | 2.12 | 1.19 | -0.49 | 0.63 |
| gehen+von | Die Tür geht von Anna kaum aufzuschließen.<br>*'The door can barely be unlocked by Anna.'* | 1.81 | 1.03 | -0.69 | 0.57 |
| gehen+von | Der Blinker geht von Younes leicht zu reparieren.<br>*'The indicator can easily be repaired by Younes.'* | 1.64 | 0.86 | -0.81 | 0.46 |
| gehört+von | Dieser laute Krach gehört von der Polizei verboten.<br>*'This loud noise ought to be forbidden by the police.'* | 3.76 | 1.25 | 0.60 | 0.64 |
| gehört-von | Dieser laute Krach gehört verboten.<br>*'This loud noise ought to be forbidden.'* | 4.01 | 1.27 | 0.76 | 0.70 |
| lassen_DO | Der Bürgermeister lässt die Bauarbeiter eine Mauer errichten.<br>*'The mayor lets the construction workers build a wall.'* | 4.67 | 0.53 | 1.19 | 0.35 |
| lassen+von | Der Bürgermeister lässt eine Mauer von den Bauarbeitern errichten.<br>*'The mayor lets a wall be built by the construction workers.'* | 4.31 | 0.86 | 0.97 | 0.54 |

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Table 10 – continued from previous page

| **Condition** | **Item (Original & Translation)** | $M$ | $SD$ | $M_z$ | $SD_z$ |
|---|---|---|---|---|---|
| middle | Das Buch liest sich gut. *'The book reads well.'* | 4.54 | 0.69 | 1.11 | 0.45 |
| middle+für | Das Buch liest sich für sie gut. *'The book reads well for her.'* | 3.65 | 1.24 | 0.54 | 0.77 |
| middle+von | Das Buch liest sich von ihr gut. *'The book reads well by her.'* | 1.47 | 0.87 | -0.92 | 0.58 |
| scheinen_past | Die Journalistin schien gut informiert zu sein. *'The journalist seemed to be well informed.'* | 4.79 | 0.41 | 1.30 | 0.23 |
| scheinen_past | Die Journalistin hat gut informiert zu sein geschienen. *'The journalist has seemed to be well informed. (compound)'* | 1.39 | 0.62 | -0.94 | 0.38 |
| sein_dat | Ihm ist nicht zu helfen. *'He cannot be helped. (sein)'* | 4.81 | 0.64 | 1.32 | 0.37 |
| sind+für | Die Fresken sind jetzt wieder für alle Besucher zu sehen. *'The frescoes can be seen again now for all visitors.'* | 4.69 | 0.52 | 1.23 | 0.40 |
| sind+von | Die Fresken sind jetzt wieder von allen Besuchern zu sehen. *'The frescoes can be seen again now by all visitors.'* | 2.88 | 1.27 | 0.03 | 0.82 |
| werden+von | Das Schmetterlingshaus wird von allen Besuchern geliebt. *'The butterfly house is loved by all visitors.'* | 4.75 | 0.55 | 1.25 | 0.46 |
| zu_inf+pl_werden | obwohl die Brücken zu reparieren versucht wurden. *'although they were attempted to repair the bridges. (PL)'* | 2.39 | 1.17 | -0.28 | 0.70 |

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Table 10 – continued from previous page

| **Condition** | **Item (Original & Translation)** | $M$ | $SD$ | $M_z$ | $SD_z$ |
|---|---|---|---|---|---|
| zu_inf+sg_werden | obwohl die Brücken zu reparieren versucht wurde.<br>*‘although it was attempted to repair the bridges. (SG)’* | 1.64 | 0.83 | -0.83 | 0.43 |